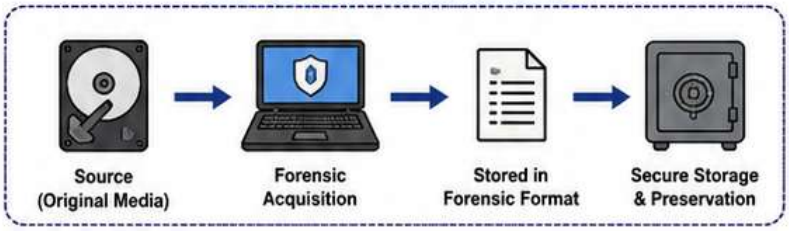
1. STORAGE FORMATS USED TO STORE COLLECTED DIGITAL EVIDENCE

INTRODUCTION

In digital forensics, after acquiring data from a storage device, the investigator needs to store the acquired data (forensic image) in a standard format. The format should preserve the integrity of the evidence and allow later verification in court.

The main storage formats are: **Raw format, Proprietary format** and **Advanced Forensic Format (AFF).**

EVIDENCE ACQUISITION AND STORAGE

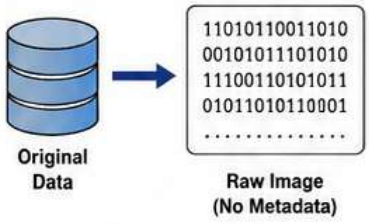


COMPARISON OF STORAGE FORMATS

Feature	Raw Format	Proprietary Format	Advanced Forensic Format (AFF)
Definition	A bit-by-bit exact copy of the original storage media without any header or metadata.	Uses the format of a specific forensic tool to store the image along with metadata.	An open, industry-standard format developed by the DFTT (Digital Forensics & Incident Response – Trusted Tools).
Structure	Only the data (bits) is copied. No information about the case, acquisition, or examiner.	Contains data + metadata in a format that can be read only by the same tool that created it.	Contains data + metadata in a standard format that can be read by multiple tools that support AFF.
Metadata	No metadata is stored.	Tool-specific metadata is stored (case info, examiner, date, hash, etc.)	Rich metadata is stored in an open and extensible manner.
Compatibility	Can be opened by any tool, but no case information is available.	Can be opened only with the tool that created the image.	Can be opened by any tool that supports AFF.
File Size	Large (only data) No additional overhead.	Large (data + metadata) Slightly larger than raw.	Large (data + metadata) Efficient and optimized.
Standard	No standard (just a stream of bits).	Proprietary to vendor (e.g., EnCase E01, FTK SMART, etc.)	Open standard (ISO/IEC 27037 supported).
Best Use	When only a simple copy is required and metadata is maintained seperately.	When using a specific forensic tool throughout the investigation.	When long-term accessibility, interoperability and standardization are required.

1. RAW FORMAT

- It is a bit-for-bit copy of the original storage media.
- Contains only data without any header, footer or metadata.
- Also called "dd image" or "flat image".
- Extension: **.dd, .raw, .bin, .img**

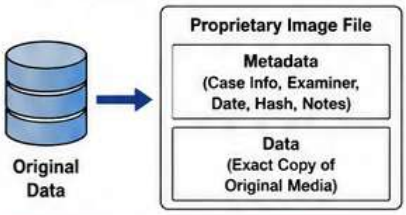


Advantages
- ✓ Simple and fast to create.
- ✓ Small overhead (only data).
- ✓ Can be opened in any forensic tool.

Disadvantages
- × No metadata or case information.
- × Difficult to prove integrity without external documentation.

2. PROPRIETARY FORMAT

- Uses the format of a specific forensic tool.
- Stores data along with metadata such as case details, examiner, date, hash value, notes, etc.
- Example formats:
  - **EnCase Evidence File (.E01)**
  - **FTK SMART (.S01)**
  - **X-Ways (.x01)**

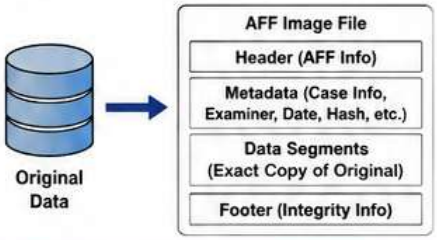


Advantages
- ✓ Stores important metadata with the image.
- ✓ Easy case management within the tool.
- ✓ Integrity verification is built-in.

Disadvantages
- × Can be opened only with the same tool.
- × Proprietary (vendor lock-in).
- × Not future proof if the tool becomes obsolete.

3. ADVANCED FORENSIC FORMAT (AFF)

- Open, standardized format developed by DFTT.
- Stores data and metadata in a structured and extensible way.
- Supported by many modern forensic tools.
- Extension: **.aff, .afm**



Advantages
- ✓ Open standard and vendor independent.
- ✓ Rich metadata and extensible.
- ✓ Compatible with multiple tools.
- ✓ Suitable for long term preservation.

Disadvantages
- × Slightly larger size due to metadata.
- × Creation may take more time than raw.

3-MARK ANSWER (KEY POINTS)

The formats used to store collected digital evidence are Raw format, Proprietary format and Advanced Forensic Format (AFF).
Raw format is a bit-for-bit copy without metadata. Proprietary format stores data along with metadata in a tool-specific format. AFF is an open standard format that stores data and metadata and can be accessed by multiple tools.

6-MARK ANSWER (STRUCTURE)

1. Introduction – Need for storage formats
2. **Raw Format** – Definition, features, advantages, disadvantages
3. **Proprietary Format** – Definition, features, advantages, disadvantages
4. **Advanced Forensic Format (AFF)** – Definition, features, advantages, disadvantages
5. Comparison Table
6. Conclusion

POINTS TO REMEMBER

- Raw – Only Data, No Metadata
- Proprietary – Data + Metadata (Tool Specific)
- AFF – Data + Metadata (Open Standard)
- AFF is the future of digital forensics.

☆ **IMPORTANT:** The main goal of any storage format is to preserve the integrity, authenticity and accessibility of digital evidence.

2. CYBER FORENSIC INVESTIGATION PROCESS

WHAT IS CYBER FORENSICS?

Cyber forensics is the process of **identifying, preserving, extracting, examining, analyzing** and **presenting** digital evidence from computers, storage media or other digital devices in a manner that is legally acceptable in a court of law.

It helps to find out what happened, how it happened, who was involved and what evidence proves it.

GOALS OF CYBER FORENSIC INVESTIGATION

- ✓ Identify the incident and scope.
- ✓ Preserve the digital evidence.
- ✓ Extract accurate and forensically sound data.
- ✓ Analyze the data to find facts and clues.
- ✓ Document the findings.
- ✓ Present the evidence in a court of law.

STEPS INVOLVED IN CYBER FORENSIC INVESTIGATION

1 IDENTIFICATION / PREPARATION

Identify the incident or crime. Determine the scope and objectives of the investigation. Collect initial information about the case.

2 PRESERVATION

Preserve the crime scene and digital evidence in its original state. Prevent alteration, modification, destruction or loss of data.

3 COLLECTION / ACQUISITION

Collect the digital evidence from the source using forensic tools and techniques. Use write blockers. Create a forensic (bit-stream) copy of the data.

4 EXAMINATION

Examine the acquired data to identify and locate relevant information. Recover deleted or hidden data if possible.

5 ANALYSIS

Analyze the examined data to find patterns, connections, and evidence. Answer the key questions: what, when, where, who, how and why.

6 DOCUMENTATION

Document every step of the investigation. Maintain logs, notes, screenshots and reports. Ensure chain of custody records are updated.

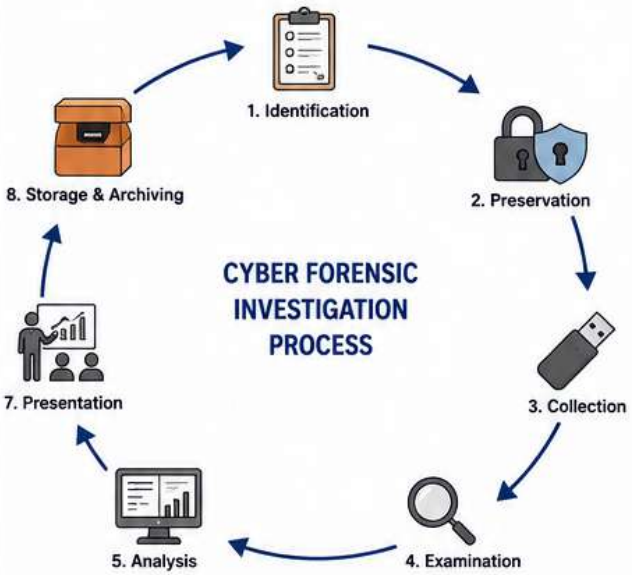
7 PRESENTATION / REPORTING

Prepare a clear and well-structured report. Present the evidence, findings, conclusions and opinions in a court of law.

8 EVIDENCE STORAGE & ARCHIVING

Store the digital evidence securely for future reference. Maintain chain of custody until the case is closed. Dispose of evidence as per legal rules after closure.

THE PROCESS IN A NUTSHELL



IMPORTANT PRINCIPLES

- ✓ **Integrity** – Evidence must be complete and unaltered.
- ✓ **Authenticity** – Evidence must be genuine.
- ✓ **Reliability** – Methods and tools used must be reliable.
- ✓ **Admissibility** – Evidence must be acceptable in court.
- ✓ **Chain of Custody** – Proper documentation of who collected, handled and transferred the evidence and when.

DETAILED EXPLANATION OF EACH STEP

1	Identification / Preparation	Receive complaint or alert. Identify the system involved. Understand the type of incident (e.g., hacking, data theft). Plan the investigation, define scope and resources.
2	Preservation	Secure the scene (physical & digital). Do not power off running systems unless necessary. Prevent remote access to the system. Maintain the current state of data.
3	Collection / Acquisition	Collect volatile data (RAM, running processes, network connections). Take forensic image using write blockers. Collect non-volatile data (HDD, SSD, mobile devices, logs).
4	Examination	Verify integrity using hash values (MD5/SHA). Recover deleted files, hidden partitions, slack space, metadata, etc.
5	Analysis	Correlate data from multiple sources. Build timelines. Identify suspicious activities, user actions, attack methods, malware, etc.
6	Documentation	Maintain case notes, logs, photos, hashes, tool details, commands used and results. Proper documentation makes the case strong in court.
7	Presentation / Reporting	Write a clear report of findings with evidence and screenshots. Explain in simple language for non-technical people (e.g., judge, lawyer).
8	Storage & Archiving	Store original evidence and copies in secure storage. Maintain chain of custody records. Keep evidence safe until the case is disposed as per legal rules.

TYPICAL INVESTIGATION FLOW (EXAMPLE)

Incident Occurs (e.g., Unauthorized access)
↓
Investigator is informed
↓
Identify incident & plan investigation
↓
Preserve digital evidence
↓
Collect data using forensic tools
↓
Examine collected data
↓
Analyze and interpret results
↓
Document all findings
↓
Prepare report
↓
Present in court
↓
Store/Archive evidence

EXAM TIP: Always mention that the process must follow legal procedures and maintain chain of custody. Use write blockers while acquiring data and calculate hash values to verify integrity.

3. CORPORATE INVESTIGATIONS

Corporate investigations are performed to detect, analyze and respond to improper or illegal use of organizational resources. The main corporate investigations include: **Internet Abuse Investigation**, **E-mail Abuse Investigation** and **Industrial Espionage Investigation**.

1. INTERNET ABUSE INVESTIGATION

Internet abuse is the improper use of the Internet facilities provided by the organization for personal gain, illegal activities or policy violations.

PROCEDURES / STEPS

1. **Receive Complaint / Detect Incident**
   Receive a complaint or detect suspicious Internet activity. Perform initial assessment.
2. **Define Scope & Objectives**
   Define the scope, time period, systems and objectives of the investigation.
3. **Obtain Authorization**
   Take written approval from management / legal team. Ensure compliance with company policy and law.
4. **Identify & Preserve Sources**
   Identify potential sources such as proxy server, firewall, router, DNS, workstation, web logs. Preserve and secure logs (use write blockers).
5. **Collect Evidence**
   Collect Internet logs, browser history, cache, cookies, DNS logs, proxy logs, firewall logs, chat logs, downloads, etc.
6. **Examine & Analyze**
   Analyze the collected data to determine the nature, extent and impact of Internet abuse. Correlate events and timeline.
7. **Document Findings**
   Document all findings with screenshots, logs and notes.
8. **Report & Take Action**
   Prepare final report and submit to management. Recommend disciplinary / legal action if required.

KEY EVIDENCE COLLECTED

- Proxy / Web server logs
- Browser history, cache, cookies
- Firewall logs
- DNS query logs
- Downloaded files
- Chat / Social media logs
- Email attachments downloaded
- System logs

2. E-MAIL ABUSE INVESTIGATION

E-mail abuse is the misuse of the organization's e-mail system for unauthorized, inappropriate or illegal purposes.

PROCEDURES / STEPS

1. **Receive Complaint / Detect Incident** r detect suspicious email activity.
2. **Define Scope &**
   Define scope: user(s), time period, mail server, and objectives.
3. **Obtain Authorization**
   Take written approval from management / legal team.
4. **Identify & Preserve Sources**
   Identify mail server, mail client, backup tapes, PST files, system logs. Preserve them securely.
5. **Collect Evidence**
   Collect emails, headers, attachments, mail server logs, sent/received items, deleted items, etc.
6. **Examine & Analyze**
   Analyze email content, headers, attachments, timestamps and routing information.
7. **Document Findings**
   Document the findings with proper evidence references and maintain chain of custody.
8. **Report & Take Action**
   Prepare and submit report. Recommend appropriate disciplinary or legal action.

KEY EVIDENCE COLLECTED

- Email messages (sent, received, drafts, deleted)
- Email headers
- Attachments
- Mail server / Exchange logs
- PST / OST files
- Backup archives
- System / Event logs

3. INDUSTRIAL ESPIONAGE INVESTIGATION

Industrial espionage is the theft of confidential information, trade secrets or intellectual property for competitive advantage or financial gain.

PROCEDURES / STEPS

A. BEFORE INVESTIGATION

1. **Determine Whether It Is Industrial Espionage**
   Verify if the incident involves theft of confidential / proprietary data.
2. **Consult Attorneys / Management**
   Consult legal team and management to understand legal implications.
3. **Determine Required Information**
   Identify what information was compromised and how it may have been stolen.
4. **Prepare Keywords / Search Terms**
   Prepare list of keywords, project codes, file names, suspect names, email IDs, etc.
5. **Gather Resources**
   Assemble investigation team, forensic tools, and technical resources.
6. **Define Goal & Scope**
   Define objectives, scope, boundaries, systems, and time frame.
7. **Obtain Management**
   Get approval to initiate formal investigation.

B. INVESTIGATION

8. **Gather Investigation Team**
   Form a team with clear roles (investigator, analyst, legal, IT).
9. **Gather Resources**
   Collect and prepare necessary forensic tools and hardware.
10. **Surveillance / Monitoring**
    Monitor suspect activities, network traffic, emails, file transfers, etc.
11. **Collect Additional Evidence**
    Collect computers, storage devices, documents, emails, logs, removable media, etc.
12. **Analyze Evidence**
    Examine collected data to identify stolen information and responsible person(s).
13. **Report to Management / Attorneys**
    Submit detailed report with findings, evidence and recommendations.

KEY EVIDENCE COLLECTED

- Confidential documents, drawings, source codes
- Emails, chat logs, voice logs
- Removable media (USB, CDs, external drives)
- Network logs, file transfer logs
- Access logs, login records
- Surveillance data, system logs

COMMON PRINCIPLES FOR ALL INVESTIGATIONS

- ✓ Follow company policy and legal guidelines.
- ✓ Preserve all evidence – do not alter or delete anything.
- ✓ Maintain chain of custody for every piece of evidence.
- ✓ Use write blockers while acquiring data.
- ✓ Document every step of the investigation.
- ✓ Ensure confidentiality and integrity of evidence.

TYPICAL FLOW



☆ **EXAM TIP:** While answering, write the steps in proper sequence and include key evidence collected in each type of investigation.

4. SECURING DIGITAL EVIDENCE

Securing digital evidence means protecting the **originality, integrity, authenticity and reliability** of the evidence from the time it is collected until it is presented in the court of law.

PROCEDURES FOR SECURING DIGITAL EVIDENCE

1. **Identify & Document**
   Identify potential digital evidence. Note the location, device type, owner, date and time. Take photographs if possible.
2. **Isolate the System**
   Disconnect the system from network (LAN, Wi-Fi, Internet) to prevent remote access or alteration of data.
3. **Preserve the Scene**
   Do not power off the system unless necessary.
   Preserve volatile data if required (RAM, running processes).
4. **Collect the Evidence**
   Use write blockers and forensic tools to acquire data.
   Create a bit-stream (forensic) copy of the storage media.
5. **Verify Integrity**
   Generate hash values (MD5 / SHA-1 / SHA-256) of the original device and the acquired image. Compare and verify.
6. **Label the Evidence**
   Assign a unique case / item number.
   Label all media, logs, photographs and documents properly.
7. **Package the Evidence**
   Place the evidence in appropriate evidence bags / containers. Seal with tamper-evident tape and sign.
8. **Complete Documentation**
   Fill the Evidence Custody Form (Chain of Custody).
   Record every transfer of the evidence.
9. **Transport the Evidence**
   Transport in a secure manner. Maintain physical control and protect from damage, moisture, shock and heat.
10. **Store the Evidence**
    Store in a secure evidence locker / room with limited access.
    Maintain environmental conditions and regular audit.

EVIDENCE PACKAGING MATERIALS



Evidence Bags
Specially designed tamper-evident bags used to secure and catalog evidence.



Anti-Static Bags / Pads
Protect electronic devices and storage media from static electricity.



Padded Containers
Prevent physical damage, shock and vibration during transportation.



Evidence Tape
Tamper-evident tape used to seal the evidence. Any tampering will be visible.



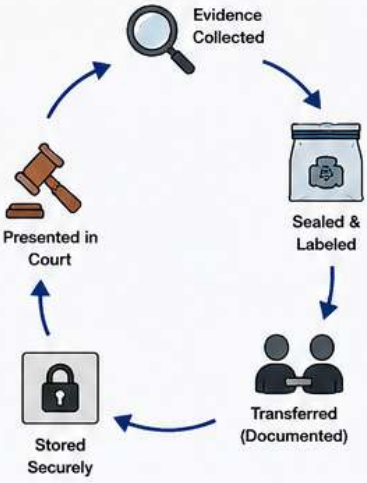
Labels / Tags
Used to uniquely identify the evidence with case number, date, time, collector etc.



Markers
Use indelible ink to write on labels, bags and containers.

CHAIN OF CUSTODY

Chain of custody is a documented and unbroken account of who handled the evidence, when, why, how and where it was transferred.



IMPORTANCE OF CHAIN OF CUSTODY

- ✓ Proves the integrity and authenticity of the evidence.
- ✓ Prevents tampering, alteration or substitution.
- ✓ Ensures the evidence is admissible in court.
- ✓ Provides a clear history of possession.
- ✓ Builds confidence in the investigation process.

EVIDENCE CUSTODY FORM (CHAIN OF CUSTODY FORM)

DESCRIPTION

The Evidence Custody Form is a legal document used to record the seizure, handling, transfer, analysis and final disposition of evidence.

PURPOSE

- Maintain continuous record of custody.
- Ensure evidence is not tampered.
- Establish who had possession and when.
- Support the admissibility of evidence in court.

EVIDENCE CUSTODY FORM

Case No.: _____ Item No.: _____ Date Collected: _____
Description of Evidence: _____
Location Found / Seized From: _____
Collected By (Name, ID): _____ Contact No.: _____
Seized From (Name): _____ Contact No.: _____

CHAIN OF CUSTODY RECORD

S. No.	Date / Time	From (Name / ID)	To (Name / ID)	Purpose / Reason	Signature
1					
2					
3					
4					
5					

Final Disposition: ☐ Returned ☐ Destroyed ☐ Retained ☐ Other (Specify) _____
Remarks: _____

INFORMATION CONTAINED

- Case / Item number
- Description of evidence
- Location of seizure
- Date and time of collection
- Collected by (name, ID, signature)
- From whom seized
- Detailed chain of custody record (date, time, from, to, purpose, signatures)
- Final disposition of evidence

TRANSPORTATION OF EVIDENCE

- Use proper packaging (evidence bags, containers).
- Seal and sign across the seal.
- Keep evidence away from extreme heat, moisture, magnetic fields and direct sunlight.
- Transport in a locked vehicle / secure box.
- Maintain the chain of custody during transportation.
- Avoid unnecessary handling.



STORAGE OF EVIDENCE

- Store in a secure evidence room / locker.
- Access only to authorized personnel.
- Maintain proper environmental conditions (temperature, humidity).
- Keep digital media in anti-static bags.
- Maintain inventory log of all stored items.
- Regularly audit and verify stored evidence.



BEST PRACTICES

- ✔ Always use write blockers during acquisition.
- ✔ Calculate and verify hash values.
- ✔ Document every action.
- ✔ Never alter original evidence.
- ✔ Follow legal guidelines and organizational policies.
- ✔ Maintain professionaliam and confidentiality.

KEY POINTS TO REMEMBER

- ★ Secure the evidence at the source.
- ★ Document everything.
- ★ Maintain an unbroken chain of custody.
- ★ Protect the evidence during transportation and storage.
- ★ Integrity of evidence = Strength of the case in court.

EXAM TIP

For 6-mark answers, explain:

- Steps for securing digital evidence
- Packaging materials
- Chain of custody
- Evidence custody form with importance
- Transportation and storage

FLOW SUMMARY



★ Remember: Integrity of evidence is the key to successful prosecution.

SURE QUESTION ★

5. AUTOPSY FORENSIC TOOL

Autopsy®
OPEN | EXTENSIBLE | FAST

WHAT IS AUTOPSY?

Autopsy is a **free, open source** digital forensics platform developed by **Basis Technology.** It is used to analyze disk images and recover evidence from computers, mobile devices and other digital media.

It has a graphical user interface and is extensible through modules and plug-ins.



KEY FEATURES OF AUTOPSY

CASE MANAGEMENT
Create, open and manage multiple forensic cases.

DATA SOURCES
Add disk images, logical files, physical devices and folders.

INGEST MODULES
Built-in modules to extract useful artifacts automatically (Ex: Recent Activity, Hash Lookup, Email, etc.).

EXAMINE EVIDENCE
View files, file metadata, hexadecimal data and thumbnails.

DELETED DATA RECOVERY
Recover deleted files and analyze unallocated space.

TIMELINE ANALYSIS
View activities in a timeline (file creation, modification, deletion, program execution, etc.).

KEYWORD SEARCH
Search for keywords in file names, file contents and metadata.

EMAIL ANALYSIS
Parse and analyze email files (PST, OST, MBOX, EML).

WEB ARTIFACTS ANALYSIS
Extract browser history, cookies, downloads, bookmarks.

HASH SET LOOKUP
Find known bad/good files using hash sets (MD5, SHA1, etc.).

REPORT GENERATION
Generate HTML reports with findings, evidence and details.

EXTENSIBLE
Supports modules and plug-ins to extend functionality.

SUPPORTED FILE SYSTEMS

- NTFS
- FAT12 / FAT16 / FAT32
- exFAT
- HFS+ (Mac)
- EXT2 / EXT3 / EXT4 (Linux)
- UFS, ISO9660, etc.

COMMON INGEST MODULES

- Recent Activity
- Hash Lookup
- Email Parser
- File Type Identification
- Keyword Search
- Picture Analyzer
- File Integrity

ADVANTAGES

- ✓ Free and open source
- ✓ User friendly GUI
- ✓ Supports many file systems
- ✓ Powerful analysis capabilities
- ✓ Extensible with plug-ins
- ✓ Generates detailed reports
- ✓ Active community support

LIMITATIONS

- ✗ Some advanced features require plug-ins
- ✗ Large cases may take more time
- ✗ Not all file systems are fully supported
- ✗ Requires good knowledge for effective analysis

APPLICATIONS / USES OF AUTOPSY

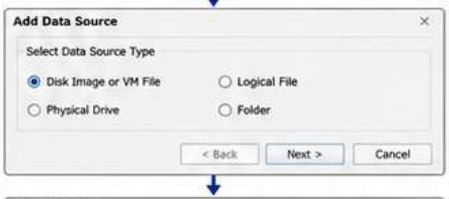
- ✔ Disk image analysis (Hard disk, SSD, USB, CD/DVD, etc.)
- ✔ File system analysis (NTFS, FAT, exFAT, HFS+, EXT, etc.)
- ✔ Recover deleted files and unallocated space analysis
- ✔ Keyword search in files and metadata
- ✔ Email analysis (PST files, EML files)
- ✔ Web history, bookmarks and cookies analysis
- ✔ Extract and analyze logs, OS artifacts and system information
- ✔ Timeline analysis of user activities
- ✔ Hash set lookups (MD5, SHA-1, SHA-256)
- ✔ Generate comprehensive forensic reports
- ✔ Support for many ingest modules and plug-ins
- ✔ Cross-platform (Windows, Linux, macOS)
- ✔ Open source and no licensing cost

WORKING / BASIC PROCESS IN AUTOPSY

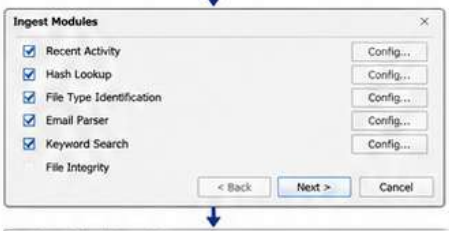
1 CREATE A CASE
Start Autopsy and create a new case. Provide case name, examiner name and case number.



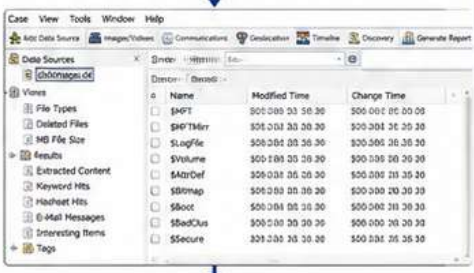
2 ADD DATA SOURCE
Add disk image file, logical file, physical device or folder as a data source.



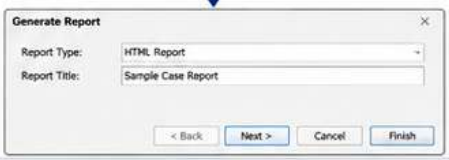
3 CONFIGURE INGEST MODULES
Select the modules you want to run during analysis.



4 ANALYZE EVIDENCE
Autopsy runs the modules and ingests data. Results are available in the case tree.



5 GENERATE REPORT
Review findings and generate a detailed HTML report.



EXAM KEY POINTS (TO REMEMBER)

- ★ Autopsy is free, open source and developed by Basis Technology.
- ★ Used for analyzing disk images and digital evidence.
- ★ Supports many file systems and data sources.
- ★ Has ingest modules for extracting useful artifacts.
- ★ Can recover deleted files and analyze unallocated space.
- ★ Provides keyword search, hash lookup, timeline and email analysis.
- ★ Generates comprehensive HTML reports.

Autopsy®

EXAM TIP: Whenever you answer about Autopsy, mention its features, modules, data sources, analysis capabilities and report generation. Diagrams / screenshots increase your answer quality.

6. AREAS OF CYBER FORENSICS

Cyber forensics is a broad field that deals with the identification, preservation, analysis and presentation of digital evidence. It has many specialized areas depending on the type of digital device, network or data source involved.

MAJOR AREAS OF CYBER FORENSICS (ANY SIX)

1. COMPUTER FORENSICS



Involves investigation of data stored on computers, laptops and storage media.

Example Evidence

- Hard disk data
- Deleted files
- System logs
- Registry files
- Application data
- Temporary files

2. NETWORK FORENSICS



Focuses on analyzing network traffic and activities to detect and investigate unauthorized access, attacks and misuse.

Example Evidence

- Network logs
- Packet captures
- Firewall logs
- IDS/IPS alerts
- Router / switch logs
- Connection records

3. MOBILE DEVICE FORENSICS



Deals with extraction and analysis of data from mobile phones, smartphones, SIM cards and tablets.

Example Evidence

- Call logs
- SMS / MMS
- Contacts
- Photos / Videos
- App data
- Location data

4. EMAIL FORENSICS



Involves analysis of email messages and email systems to detect abuse, fraud or policy violations.

Example Evidence

- Email headers
- Email content
- Attachments
- Sent / received logs
- Mail server logs
- Deleted emails

5. WEB FORENSICS (ONLINE FORENSICS)



Covers investigation of web related activities performed by users on the internet.

Example Evidence

- Browser history
- Cookies
- Cache files
- Downloaded files
- Bookmarks
- Web server logs

6. DATABASE FORENSICS



Involves extraction and analysis of data from databases to detect tampering, fraud or data leakage.

Example Evidence

- Database files
- Query logs
- Transaction logs
- User activity logs
- Backup files
- Deleted records

7. CLOUD FORENSICS



Deals with investigation of data, logs and activities in cloud computing environments such as SaaS, PaaS and IaaS.

Example Evidence

- Cloud storage data
- Access logs
- Virtual machine logs
- Activity logs
- API logs
- Deleted / recovered data

8. DIGITAL FORENSICS (GENERAL)



A broad term that includes all techniques used to identify, collect, analyze and present digital evidence from any digital source.

Example Evidence

- Documents
- Images
- Audio / Video
- System files
- User activity
- Metadata
- Log files
- Internet artifacts

KEY POINTS

- ✓ Each area has specific tools, techniques and procedures.
- ✓ The goal is to collect reliable digital evidence without altering the original data.
- ✓ Evidence collected from any of these areas must be admissible in court.

RELATIONSHIP BETWEEN AREAS

All these areas are interconnected. For example, an investigation may start with web forensics, lead to email forensics, involve mobile device forensics and finally find evidence in a database or cloud environment.



IMPORTANCE

- ✦ Helps in detecting and investigating cyber crimes.
- ✦ Supports incident response and damage assessment.
- ✦ Provides digital evidence for legal action and prosecution.

EXAM TIP: If asked to "List any six areas of cyber forensics", write the names and a one line description with an example. Diagrams or icons increase your score.

7. COMPUTER CRIME



1. WHAT IS COMPUTER CRIME?

Computer crime is any illegal activity in which a computer is the **object**, the **tool**, or the **place** of the offence.

- It involves the use of computers, networks or digital devices to commit unlawful acts, or the targeting of computer systems and data.
- These crimes violate laws and regulations and cause harm to individuals, organizations or the society.

2. GENERAL CATEGORIES OF COMPUTER CRIME

Computer crimes can be broadly classified into the following categories:

Category	Description	Examples	Icon
1. Unauthorized Access	Gaining access to a computer system or data without permission.	• Hacking • Password cracking • Bypassing security systems	
2. Illegal Interception	Intercepting or monitoring data, communication or information without authorization.	• Packet sniffing • Eavesdropping on emails • Wiretapping	
3. Data Interference	Altering, damaging, deleting, or suppressing computer data or programs without authority.	• Data modification • Malware attacks • Ransomware	
4. System Interference	Damaging or disrupting the normal functioning of computer systems or networks.	• Denial of Service (DoS) • Distributed DoS (DDoS) • Virus and worm attacks	
5. Misuse of Devices	Using computer devices or resources for unlawful purposes.	• Spamming • Phishing • Identity theft	
6. Content Related Crimes	Creating, publishing or distributing illegal or obscene digital content.	• Child pornography • Hate speech • Cyber defamation	
7. Cyber Terrorism	Using computers or networks to threaten or intimidate individuals, organizations or governments.	• Threatening attacks • Hacking critical infrastructure • Spreading terror messages	
8. Intellectual Property Crimes	Violating copyright, trademark or license protected by law.	• Software piracy • Illegal copying • Copyright infringement	

3. HOW IS CYBERCRIME RELATED TO CYBER FORENSICS?

- Cyber forensics is the science of identifying, preserving, analyzing and presenting digital evidence in a way that is legally acceptable.
- Cybercrime and cyber forensics are closely related as one deals with the crime, and the other with the investigation.
- Whenever a cyber crime occurs, digital evidence is created, stored or transmitted in some form. Cyber forensics helps to collect and analyze this evidence to find:
  - ❖ What happened?
  - ❖ How it happened?
  - ❖ Who is responsible?
  - ❖ What is the impact?
- The findings of cyber forensics help in taking legal action against the offender.
- Without cyber forensics, it is difficult to prove cybercrimes in court.

RELATIONSHIP BETWEEN CYBERCRIME AND CYBER FORENSICS



CYBERCRIME	DIGITAL EVIDENCE	CYBER FORENSICS	LEGAL ACTION
Illegal activity occurs	is generated (logs, files, emails, metadata, etc.)	collects, preserves and analyzes evidence	Evidence is presented in court and used for prosecution

- ✓ Cyber forensics turns digital evidence into legal proof.
- ✓ It supports law enforcement agencies in investigating and solving cybercrimes.
- ✓ It helps organizations in incident response and prevention of future attacks.

IMPORTANCE



- ➢ Helps in detecting and understanding cybercrimes.
- ➢ Identifies the source, method and impact of the attack.
- ➢ Provides admissible evidence for legal proceedings.
- ➢ Assists in recovering lost or stolen data.
- ➢ Helps in improving security measures and preventing future crimes.

EXAMPLES OF COMPUTER CRIMES

- An employee stealing company data and selling it to competitors.
- Hacking into a bank's server and transferring money.
- Spreading malware that damages systems.
- Sending fake emails to steal user credentials (phishing).
- Uploading copyrighted movies or software illegally.



EXAM TIP: **For 3 marks:** Define computer crime and list its categories.
For 6 marks: Explain computer crime, its categories and the relationship with cyber forensics.

8. COMPANY POLICY VIOLATION

Company policy violation refers to any action by an employee or user that goes against the rules, acceptable use policies, or standard operating procedures defined by the organization.

1. ACTIVITIES CONSIDERED AS COMPANY POLICY VIOLATIONS

The following activities are commonly treated as company policy violations:

Category	Activities (Examples)	Purpose of Policy
Internet Misuse	• Visiting inappropriate websites • Streaming videos or online games during working hours • Downloading unauthorized software or files	To ensure productivity, prevent malware infections and reduce bandwidth misuse.
Email Misuse	• Sending or receiving inappropriate or offensive emails • Sharing confidential information • Using company email for personal communication	To maintain professional communication and protect confidential information.
Unauthorized Use of Devices / Resources	• Using USB drives or external devices without permission • Installing unauthorized hardware • Using company resources for personal benefit	To prevent data leakage, malware infections and ensure proper use of company assets.
Security Violations	• Sharing passwords • Bypassing security controls • Disabling antivirus or firewall • Attempting unauthorized access	To protect systems, networks and data from unauthorized access and attacks.
Data Handling Violations	• Accessing data without authorization • Modifying or deleting important data • Failing to follow data backup procedures	To ensure data integrity, availability and compliance with legal requirements.
Work Behaviour Violations	• Harassment or bullying via digital channels • Sharing offensive content • Using company systems for illegal activities	To maintain a safe, ethical and respectful work environment.

Note: Company policies are usually documented in the Acceptable Use Policy (AUP) or Employee Handbook. Violations may lead to disciplinary action or legal action.

2. STEPS INVOLVED IN INVESTIGATING A COMPANY POLICY VIOLATION

1 Receive Complaint / Report
A complaint is received from management, HR, or through monitoring tools about a possible policy violation. Example: HR reports an employee is downloading large personal files during work hours.

2 Preliminary Assessment
Verify the complaint and determine if it falls under policy violation. Define the scope, objectives and systems involved.

3 Obtain Authorization
Get written permission from management or legal authority to start the investigation. Ensure compliance with company policy and local laws.

4 Identify and Preserve Evidence
Identify potential sources of evidence such as computers, email servers, logs, network devices etc. Preserve the evidence using proper techniques (e.g., imaging, log collection). Maintain chain of custody.

5 Collect Evidence
Collect relevant digital evidence such as log files, email records, internet history, accessed files, copied data, USB usage logs, etc.

6 Examine and Analyze
Analyze the collected evidence using forensic tools and techniques to determine what happened, how it happened, when it happened and who was involved.

7 Document Findings
Prepare a detailed investigation report including methods used, evidence collected, analysis results and conclusions. Attach relevant logs, screenshots and supporting documents.

8 Take Action
Submit the report to management / HR. Based on the findings, take appropriate action such as warning, suspension, training or legal action.

9 Review and Improve Policies
Review the policy and security controls to prevent similar violations in the future. Provide employee awareness and training if required.

EXAMPLE

Scenario: An employee is suspected of transferring confidential company files to a personal USB drive.

Investigation Process:

- HR reports unusual file copy activity.
- Investigator gets authorization from management.
- Logs from the endpoint security tool and USB device history are collected.
- System is imaged and USB usage logs are preserved.
- Analysis shows the employee copied 500 MB of confidential files to a USB at 3:15 PM on a working day.
- Report is generated and submitted.
- Employee is warned and given security awareness training.
- Policy is updated to restrict USB usage by default.

3. RELATIONSHIP BETWEEN COMPANY POLICY VIOLATION AND CYBER FORENSICS

Cyber forensics plays a vital role in investigating company policy violations.

It provides:

- ✓ Digital evidence that proves or disproves the violation.
- ✓ Objective analysis of what happened and who is responsible.
- ✓ Support for disciplinary action while ensuring legal compliance.
- ✓ Helps improve policies and security controls.

KEY POINTS

- Company policy violations reduce productivity, compromise security and cause financial loss.
- Every organization must have clear policies and monitor compliance.
- Digital evidence and proper investigation ensure fairness and support legal action.
- Maintain chain of custody, document every step and follow legal guidelines.

IMPORTANCE

- ★ Protects company data and assets.
- ★ Ensures employee accountability.
- ★ Helps in identifying insider threats.
- ★ Supports legal proceedings if required.

EXAM TIP

For 3 marks: List any activities considered as policy violations.
For 6 marks: Explain the investigation steps with an example.

9. BIT-STREAM / FORENSIC COPY

In digital forensics, an exact and unaltered copy of digital data is required for investigation. This is achieved using **Bit-stream (Forensic) Copy** or **Sector Copy**.

1. WHAT IS A FORENSIC COPY?

A forensic copy is an exact, bit-by-bit duplicate of the original data source (entire drive or selected media) that preserves all data, including deleted files, hidden data, slack space, file system structures and metadata.

- It is used for examination so that the original evidence remains untouched and its integrity is maintained.
- It is also called an Evidence Copy.

2. TYPES OF COPYING METHODS

- **Normal Copy** – Copies only the files that are visible and accessible to the user.
- **Sector Copy** – Copies data sector by sector (512 bytes/sector) from the source.
- **Bit-stream (Forensic) Copy** – Creates a bit-for-bit image of the entire source, including all sectors and metadata.

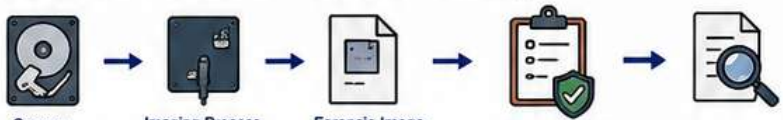
3. DIFFERENCE BETWEEN NORMAL COPY, FORENSIC (BIT-STREAM) COPY AND SECTOR COPY

Feature	Normal Copy	Sector Copy	Bit-stream / Forensic Copy
Definition	Copies only selected files or folders from source to destination.	Copies data sector by sector from the source.	Creates an exact bit-by-bit image of the entire source including all sectors.
Data Copied	Only allocated files and visible data.	All sectors (allocated + unallocated) but may skip some metadata.	All data including allocated, unallocated, slack space, metadata, hidden data, file system structures.
Deleted Data Recovery	Deleted data is not copied.	Possible (if sectors are not overwritten).	All deleted data can be recovered.
File System Information	File system metadata not preserved completely.	Partially preserved (some tools may not copy all metadata).	Completely preserved including boot sector, MBR/GPT, timestamps, etc.
Integrity / Exactness	Not exact. Only duplicates user-accessible data.	Exact at sector level but may miss certain hidden or file system structures.	100% exact replica of the source.
Use in Forensics	Not suitable for forensic analysis.	Suitable but not as complete and reliable as bit-stream copy.	Most preferred method in forensic investigations.
Tools Used	File explorer, drag & drop, copy command.	dd, DCFLDD, FTK Imager (sector copy option), OSFClone.	EnCase, FTK Imager, dd (with hash), X-Ways, Guymager, OSFClone.
Hash / Verification	No hash verification.	Hash can be generated for verification.	Hash is generated (MD5, SHA-1/SHA-256) to ensure integrity.
Time Taken	Fast.	Medium.	Longer (copies entire drive).
Example	Copying a document from USB to computer.	Imaging a hard disk using dd command.	Creating an E01 image of a drive using FTK Imager.

4. BIT-STREAM (FORENSIC) COPY – EXPLANATION

- Bit-stream copying reads every single bit from the source media and writes it to another file (image file) or another drive.
- The copy includes:
  - Allocated data (existing files)
  - Unallocated space
  - Deleted files and fragments
  - Slack space
  - File system metadata
  - Hidden partitions and system areas
- A hash value is generated for both source and image to verify that the copy is exact and untampered.

5. HOW FORENSIC COPY IS CREATED (BIT-STREAM COPY)



Source (Original Drive) – Evidence is identified.

Imaging Process (Using Forensic Tool) – Bit-by-bit reading of source drive.

Forensic Image (E01 / DD / RAW) – Exact replica created and saved as image file.

Hash Verification – Hash values are generated and matched to ensure integrity.

Analysis – Image is examined. Original evidence remains untouched.

Common Hash Functions: MD5, SHA-1, SHA-256

Purpose: To verify that the image is identical to the original source.

6. SECTOR COPY – EXPLANATION

- Sector copy reads data in blocks (sectors) from the source and copies to destination.
- Usually copies all sectors including unallocated space.
- Some tools may skip bad sectors or hidden areas depending on settings.
- Less comprehensive than bit-stream copy but still useful.

7. NORMAL COPY – EXPLANATION

- Only files/folders that are visible are copied.
- Does not copy deleted files, slack space or metadata.
- Not acceptable in forensic investigations as it may alter timestamps and properties.

8. ADVANTAGES OF BIT-STREAM COPY

- ✓ Exact and complete replica
- ✓ Ensures data integrity
- ✓ Preserves deleted/hidden data
- ✓ Admissible in court
- ✓ Original evidence is protected

9. LIMITATIONS

- × Time consuming
- × Requires large storage space
- × Special tools and skills required

10. KEY POINTS

- ★ Always work on the forensic copy, never on the original evidence.
- ★ Generate and verify hash values before and after copying.
- ★ Maintain chain of custody for the evidence.
- ★ Document every step of the imaging process.

11. EXAM TIP

For 3 marks: Define forensic copy and list the types.

For 6 marks: Explain bit-stream copy and differentiate between normal copy, sector copy and bit-stream copy with points.

10. NTFS (NEW TECHNOLOGY FILE SYSTEM)

1. INTRODUCTION TO NTFS

- NTFS (New Technology File System) is the standard file system used by Windows NT family of operating systems.
- It was introduced with Windows NT 3.1 in 1993 and has been improved in later versions (Windows 2000, XP, Vista, 7, 8, 10, 11).
- It is a journaling file system that provides high reliability, security, large storage support and advanced features compared to FAT.
- NTFS is widely used in forensics because of its robustness and the rich metadata it maintains.

2. KEY FEATURES OF NTFS

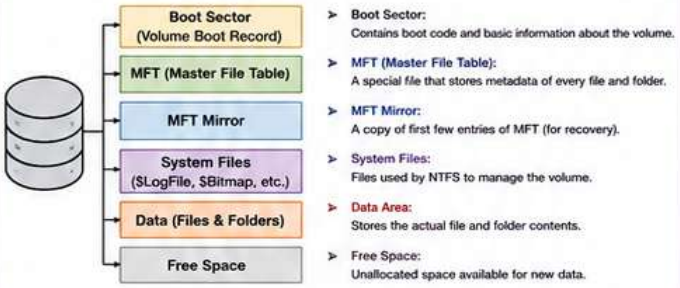
- ✓ Supports very large files (up to 16 EB) and large volumes (up to 256 TB).
- ✓ Uses smaller cluster size (from 512 bytes to 64 KB).
- ✓ Provides file and folder permissions (Security Descriptor).
- ✓ Supports encryption (EFS – Encrypting File System).
- ✓ Supports compression to save disk space.
- ✓ Supports hard links, symbolic links and junctions.
- ✓ Includes auditing and access control.
- ✓ Journaling (via $LogFile) ensures reliability and quick recovery.
- ✓ Built-in disk quota management.
- ✓ Better performance and security than FAT.

3. WHERE IS NTFS USED?

- Internal hard disks and SSDs in Windows systems.
- External drives, USB drives, memory cards (via formatting).
- Network drives and servers.
- Preferred file system for system/reserved partitions.

4. NTFS ARCHITECTURE

NTFS is based on a Master File Table (MFT), which holds all the information about files and folders.



- **Boot Sector:** Contains boot code and basic information about the volume.
- **MFT (Master File Table):** A special file that stores metadata of every file and folder.
- **MFT Mirror:** A copy of first few entries of MFT (for recovery).
- **System Files:** Files used by NTFS to manage the volume.
- **Data Area:** Stores the actual file and folder contents.
- **Free Space:** Unallocated space available for new data.

5. MASTER FILE TABLE (MFT)

- The heart of NTFS.
- A database file that contains a record (entry) for every file and folder on the volume.
- Every MFT record has a unique MFT Reference Number.
- Default size of an MFT record is 1 KB.

Each MFT Record contains:

- ❖ Standard Information (timestamps, attributes)
- ❖ File Name Information
- ❖ Security Descriptor
- ❖ Attribute List
- ❖ Data Attributes (resident or non-resident)

Example MFT Record Layout	
0 – 3	Signature (FILE)
4 – 5	Update Sequence Array Offset
6 – 7	Update Sequence Array Size
8 – 15	Log File Sequence Number
16 – 17	Sequence Number
18 – 19	Hard Link Count
20 – 21	First Attribute Offset
22 – 23	Flags
24 – 27	Used Size of Entry
28 – 31	Allocated Size of Entry
32 – ...	**Attributes**

6. MAJOR NTFS SYSTEM FILES AND THEIR PURPOSE

System File	Description	Purpose
$MFT	Master File Table	Contains metadata of all files and folders.
$MFTMirr	MFT Mirror	Mirror of first few records of $MFT for recovery.
$LogFile	NTFS Log File (Journal)	Records file system changes before they are written to disk.
$Bitmap	Cluster Bitmap	Tracks free and used clusters on the volume.
$Boot	Boot Sector File	Contains boot sector information.
$AttrDef	Attribute Definition File	Defines the types of attributes used in MFT.
$Volume	Volume Information	Stores volume label, version and other information.
$Secure	Security Descriptor Stream	Stores security descriptors for files and folders.
$UpCase	Upper Case Table	Used for case-insensitive filename comparison.

7. NTFS ATTRIBUTES (Common Attribute Types)

Attribute ID	Attribute Name	Description
$STANDARD_INFORMATION (0x10)		Stores timestamps and file attributes (read-only, hidden, system, etc.)
$ATTRIBUTE_LIST (0x20)		Points to additional attributes if they do not fit in the MFT record.
$FILE_NAME (0x30)		Stores file name and its parent folder reference.
$DATA (0x80)		Contains the actual file or folder data.
$INDEX_ROOT (0x90)		Root of the B+ tree used for indexing in a directory.
$INDEX_ALLOCATION (0xA0)		Stores index blocks for directory contents.
$BITMAP (0xB0)		Used in $Bitmap file to mark free/used clusters.

8. NTFS DATA STRUCTURE (Simplified View)

Files are stored in clusters. A file can be:

- **Resident** – small files stored inside the MFT record itself.
- **Non-resident** – large files stored in data area, MFT record contains pointers.

Non-resident Attribute Example



9. NTFS PERMISSIONS (Security)

- Each file and folder has a Security Descriptor.
- Includes:
  - Owner
  - Group
  - Discretionary Access Control List (DACL) – allow/deny permissions
  - System Access Control List (SACL) – auditing information

Common Permissions

- ☑ Full Control
- ☑ Modify
- ☑ Read & Execute
- ☑ List Folder Contents
- ☑ Read
- ☑ Write

10. NTFS vs FAT FILE SYSTEMS

Feature	NTFS	FAT16 / FAT32 (File Allocation Table)
Full Form	New Technology File System	File Allocation Table
Introduced	1993 (Windows NT 3.1)	FAT16 (1980s), FAT32 (1996)
Maximum File Size	Up to 16 Exabytes (16 EB)	FAT16: 2 GB FAT32: 4 GB
Maximum Volume Size	Up to 256 TB (Windows) Up to 16 EB (theoretical)	FAT16: 2 GB FAT32: 2 TB
Cluster Size	512 B to 64 KB	FAT16: 2 KB to 32 KB FAT32: 4 KB to 32 KB
Security	Yes (Permissions, Encryption, Auditing)	No security (no permissions)
Compression	Yes	No
Encryption	Yes (EFS)	No
Journaling	Yes ($LogFile)	No
Reliablity	High (Journaling, Self-healing)	Low (More prone to corruption)
Fragmentation	Low (Better management)	High (in FAT32 due to design)
Metadata Storage	MFT (centralized)	FAT Tables (decentralized)
Speed	Faster and efficient	Slower for large files/volumes
Best Use	Modern OS, Servers, Large storage, Security	Older systems, Removable media, Compatibility

11. NTFS IN FORENSICS (Why Important?)

- NTFS maintains detailed metadata (timestamps, permissions, MAC times: Modified, Accessed, Created).
- $LogFile helps in timeline analysis and detecting file system changes.
- Deleted files can be recovered from MFT entries and unallocated space.
- Slack space, alternate data streams (ADS), and transaction logs may contain important evidence.
- Tools like Autopsy, FTK Imager, EnCase, X-Ways, etc., parse NTFS structures for forensic analysis.

Key Forensic Artifacts in NTFS

- $MFT – File/folder metadata
- $LogFile – File system activities
- $UsnJrnl – Change journal (on NTFS 3.0+)
- $Recycle.Bin – Deleted file information
- Timestamps – MAC times
- Slack Space – Residual data
- Alternate Data Streams (ADS)
- Unallocated Space
- Volume Shadow Copies (VSS)

KEY POINTS

- ★ NTFS is a secure, reliable and feature-rich file system.
- ★ It uses MFT to store all file and folder information.
- ★ Supports permissions, encryption, compression and journaling.
- ★ FAT file systems are simpler but lack security and reliability.

EXAM TIP

For 5 marks: Explain NTFS with architecture, MFT, system files and features.

For 2 marks: List key features or compare NTFS with FAT.

IMPORTANT

- ➤ Always write difference between NTFS and FAT if asked.
- ➤ Diagrams of disk layout and MFT structure score more.
- ➤ Mention forensic importance of NTFS.

11. FAT vs NTFS

(DIFFERENCE BETWEEN FAT AND NTFS FILE SYSTEMS)

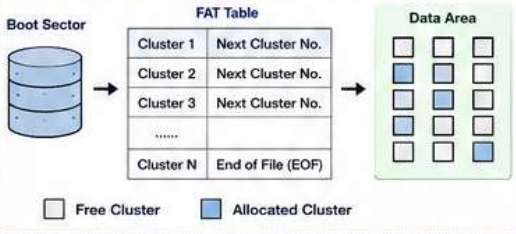
A **File System** is a way the operating system organizes and stores data on a storage device.
The two most common file systems in Windows environments are **FAT** (File Allocation Table) and **NTFS** (New Technology File System).

Feature	FAT (FAT12 / FAT16 / FAT32)	NTFS (New Technology File System)
1. Full Form	File Allocation Table	New Technology File System
2. Introduced	FAT12 (1977), FAT16 (1988), FAT32 (1996)	Windows NT 3.1 (1993)
3. Maximum File Size	FAT12: 16 MB FAT16: 2 GB FAT32: 4 GB (minus 1 byte)	Up to 16 EB (Exabytes) (16 EB = 16,384 Petabytes)
4. Maximum Volume Size	FAT12: 32 MB FAT16: 2 GB FAT32: 2 TB	Up to 256 TB (Windows) Up to 16 EB (theoretical)
5. Cluster / Allocation Unit Size	512 Bytes to 32 KB (depends on volume size)	512 Bytes to 64 KB (depends on volume size)
6. Structure / Architecture	Uses a simple File Allocation Table (one FAT for FAT12/16, two copies for FAT32)	Uses Master File Table (MFT) – a database of files and folders
7. Metadata Storage	Stored in boot sector and FAT	Stored in MFT (detailed metadata for every file and folder)
8. Security / Permissions	No built-in security (No permissions)	Supports file & folder permissions, ACLs (Access Control Lists), ownership
9. Encryption	No	Yes (EFS – Encrypting File System)
10. Compression	No	Yes (Built-in file and folder compression)
11. Journaling / Logging	No (More prone to corruption)	Yes (Journaling with $LogFile) (Better reliability and recovery)
12. Reliability	Lower (data loss possible if system crashes or power fails)	Higher (Journal and metadata redundancy reduce corruption)
13. Fragmentation	More prone to fragmentation	Less prone to fragmentation (uses extent-based allocation)
14. Filename Length	8.3 format (8 characters name, 3 characters extension)	Up to 255 Unicode characters
15. Case Sensitivity	Not case sensitive	Preserves case but not case sensitive by default
16. Supported Operating Systems	Windows, DOS, macOS, Linux, Android, Cameras, Game Consoles, etc. (Widely compatible)	Windows (Vista and later full support), Linux (read/write), macOS (read-only by default)
17. Best Use	Removable drives, memory cards, small volumes, maximum compatibility	Internal hard disks/SSDs, system drives, large files, security, performance
18. Speed / Performance	Faster for small files and on small volumes	Better performance on large volumes and with large files
19. Bad Sector Handling	Limited	Better (marks bad clusters and moves data)
20. Advanced Features	None	Hard Links, Soft Links, Alternate Data Streams (ADS), Disk Quotas, Reparse Points, Sparse Files, VSS, etc.
21. Typical Usage Today	Rarely used except for compatibility	Standard file system for modern Windows systems

ABOUT FAT FILE SYSTEM

- FAT is an older file system design.
- Very simple and widely supported.
- Lack of security, large file size limitation, and no journaling makes it less reliable.
- Still used in USB drives, SD cards, digital cameras, consoles, and embedded devices.

FAT FILE SYSTEM STRUCTURE (Simplified)



ABOUT NTFS FILE SYSTEM

- NTFS is the advanced, secure and reliable file system.
- Designed for performance, large storage, security and data integrity.
- Supports many advanced features not available in FAT.
- Default choice for Windows OS.

KEY TAKEAWAYS

- ✓ FAT = Simple, compatible, small volumes, no security, no journaling.
- ✓ NTFS = Secure, reliable, supports large files/volumes, journaling, permissions, encryption.
- ✓ Use FAT for removable devices and maximum compatibility.
- ✓ Use NTFS for internal drives and when security, reliability and advanced features are required.

EXAM TIP

For 6 Marks: Write a detailed comparison table with at least 15–20 points.

For 3 Marks: Write short points covering major differences.

NOTE: Both FAT and NTFS have their own advantages and limitations.
Choose the file system based on requirement of compatibility, security, reliability and performance.

12. NTFS HIDDEN DATA / ADS

How do you analyse hidden data in NTFS?

1. ALTERNATE DATA STREAM (ADS) IN NTFS

NTFS supports a feature called **Alternate Data Streams (ADS)**, which allows files to have multiple streams of data identified by a stream name.

- ✓ Every file has at least one stream – the default data stream (unnamed), which you see in Windows.
- ✓ ADS allows additional, hidden streams to be attached to a file without changing the file size or timestamp.
- ✓ These streams are not visible in normal file browsing and can be used to hide data.
- ✓ Attackers and malware often abuse ADS to hide malicious code, documents, scripts, or data.

Example:

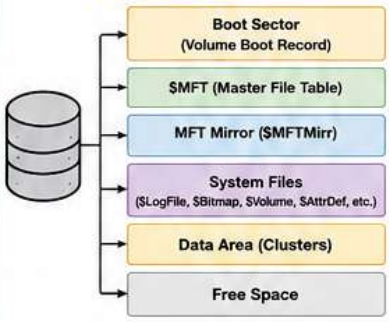
report.docx	→ Default data stream (visible)
report.docx:secret.txt	→ Alternate data stream (hidden)
photo.jpg:payload.exe	→ Alternate data stream (hidden)

ADS exists only on NTFS volumes.
It is not supported in FAT/FAT32/exFAT.

2. NTFS FILE SYSTEM OVERVIEW

- ✓ NTFS (New Technology File System) is a journaling file system used by Windows.
- ✓ All files and folders are represented in the Master File Table (MFT).
- ✓ Each entry in MFT contains metadata about a file/folder and information about where its data is stored on disk.
- ✓ NTFS has rich metadata, file attributes, security descriptors, timestamps, and supports features like compression, encryption, hard links, and ADS.

NTFS Architecture



3. MASTER FILE TABLE (MFT)

MFT is the heart of NTFS. It is a special file ($MFT) that contains a record for every file and directory.

- ❖ Each MFT record is typically 1 KB (can be 2 KB or 4 KB).
- ❖ The first 16 records are reserved for system use.
- ❖ Record Number = Reference Number.
- ❖ It stores both metadata (in the record) and, if small enough, the actual file data (resident data).

MFT Record Structure

Offset (Bytes)	Content
0 – 3	Signature ("FILE")
4 – 5	Update Sequence Array Offset
6 – 7	Update Sequence Array Size
8 – 15	Log File Sequence Number
16 – 17	Sequence Number
18 – 19	Hard Link Count
20 – 21	First Attribute Offset
22 – n	Attributes (variable length)
...	$STANDARD_INFORMATION, $FILE_NAME, $DATA, $ATTRIBUTE_LIST, $SECURITY_DESCRIPTOR, etc.

4. NTFS FILE ATTRIBUTES (Important)

Attribute	ID (Hex)	Description
$STANDARD_INFORMATION	0x10	Contains timestamps, file attributes (Read-only, Hidden, System, Archive, etc.), owner ID, security ID.
$FILE_NAME	0x30	Stores file name(s), parent directory reference, timestamps, flags.
$DATA	0x80	Contains the file data. Can be resident (inside MFT) or non-resident (in clusters). Includes ADS data.
$ATTRIBUTE_LIST	0x20	Used when a record has many attributes.
$SECURITY_DESCRIPTOR	0x90	Stores security information (DACL, SACL, owner, group).
$VOLUME_NAME	0x60	Volume label.
$LOGGED_UTILITY_STREAM ($LogFile)	0x100	Signifies NTFS journal.
$BITMAP	0x6	Tracks used and free clusters.
$VOLUME	0x3	Stores volume information.
$INDEX_ROOT / $INDEX_ALLOCATION	0x90 / 0xA0	Used for directories (B+ Tree index).

Important: The $DATA attribute can contain multiple data streams. The unnamed stream is the default data. Named streams are ADS.

5. FILE DATA AND HIDDEN DATA IN NTFS

Visible Data

- ✓ Stored in the default unnamed $DATA stream.
- ✓ Accessible through normal OS commands and file explorers.
- ✓ Example: file.txt → contains normal text.

Hidden Data

- ✓ Stored in Alternate Data Streams (ADS).
- ✓ Not visible in normal directory listing.
- ✓ Can contain any type of data.
- ✓ Example: file.txt:secret → hidden content.

Where is hidden data stored?

Hidden data (ADS) is stored in the $DATA attribute of the file's MFT record under a separate stream name.

The $FILE_NAME attribute may also indicate the presence of multiple streams.

Examples of hidden data

- ❖ Malware payloads
- ❖ Scripts, commands
- ❖ Confidential documents
- ❖ Configuration data
- ❖ Logs, tokens, keys, etc.

6. HOW TO ANALYSE HIDDEN DATA IN NTFS

Step-by-Step Procedure

1. **Identify the NTFS Volume**
   Confirm the drive is NTFS (using fsutil or file system tools).
2. **Acquire Forensic Image**
   Create a bit-stream/forensic image of the drive to preserve evidence (e.g., using FTK Imager, dd, EnCase).
3. **Parse NTFS Structures**
   Parse the image to read NTFS metadata (MFT, $LogFile, $Bitmap, etc.) using forensic tools.
4. **Extract MFT Records**
   Examine MFT entries for files of interest.
5. **Check $DATA Attribute for Additional Streams**
   Look for multiple $DATA attributes in the record. Named $DATA attributes indicate ADS.
6. **Enumerate ADS**
   List all alternate data streams for files.
7. **Extract ADS Content**
   Export and analyse the content of ADS.
8. **Analyse and Correlate**
   Correlate findings with timelines, logs, and other artefacts.

Common Forensic Tools

- Sysinternals: streams.exe, dir /R
- FTK Imager, EnCase, X-Ways Forensics
- Autopsy, Sleuth Kit (fls, icat), Bulk Extractor

7. DETECTING ADS (Common Methods)

Method	Command / Tool
Using CMD	dir /R
Using PowerShell	Get-Item -Path . -Stream *
Using Sysinternals	streams.exe -s <file>
Using FSUTIL	fsutil streams <file>
Using Forensic Tools (GUI)	Autopsy, FTK, EnCase, X-Ways

Example (CMD)

```
C:\> echo This is secret data > file.txt:secret.txt.
C:\> dir
file.txt                10,240 bytes    (visible)
C:\> dir /R
file.txt                10,240 bytes
:secret.txt:$DATA           15 bytes    (hidden stream)
```

8. PRACTICAL EXAMPLE

Suppose we create a hidden stream.

Command:

```
echo confidential > report.docx:secret.txt
```

What happens?

- report.docx is the main file (visible).
- secret.txt is hidden inside report.docx as an ADS.
- File size of report.docx remains the same.
- Not visible in normal view.

To view:

```
dir /R
```

To extract:

```
more < report.docx:secret.txt
```

To delete:

```
del report.docx:secret.txt
```

9. OTHER SOURCES OF HIDDEN DATA IN NTFS

- ❖ Unallocated Space – deleted files, old versions.
- ❖ Slack Space – residual data at the end of clusters.
- ❖ $LogFile (Journal) – past file system transactions.
- ❖ Thumbnail Cache – previews of images/documents.
- ❖ Reparse Points / Junctions – hidden links.
- ❖ Encrypted/Compressed Files – data not directly visible.
- ❖ Shadow Copies (VSS) – previous versions of files.
- ❖ Browser/Application Artifacts – stored in hidden folders.
- ❖ USN Journal – record of file changes and activities.

10. KEY POINTS

- ✓ ADS is a powerful feature of NTFS that can be misused to hide data.
- ✓ All file metadata (including ADS) is recorded in the MFT.
- ✓ Always analyse ADS during forensic investigations.
- ✓ Hidden data can be critical evidence in cybercrime cases.
- ✓ Use a combination of manual commands and forensic tools.

11. IMPORTANCE IN FORENSICS

- ✓ Helps in uncovering hidden malware, backdoors, stolen data.
- ✓ Provides visibility into attacker's attempt to hide traces.
- ✓ Crucial for insider threat, APTs, and data exfiltration cases.
- ✓ Can reveal hidden communications, credentials, and logs.

EXAM TIP

For 6 Marks:

1. Define ADS and NTFS.
2. Explain MFT, attributes, file data.
3. Explain hidden data types.
4. Explain steps to analyse hidden data.
5. Mention tools and examples.

SUMMARY: NTFS stores rich metadata in MFT. Files can have multiple data streams (ADS). Hidden data in ADS and other areas can be extracted and analysed using forensic techniques and tools.

13. WINDOWS REGISTRY

Explain the importance/usage of Windows Registry in forensic analysis.
Illustrate Windows Registry organization and functions of Registry HKEYs.
Mention any six Windows Registry terminology.

1. INTRODUCTION

The Windows Registry is a hierarchical database used by Microsoft Windows to store low-level settings for the operating system, installed applications, hardware, user preferences, and system configuration.

It acts as the central repository for configuration information. Changes in the Registry directly affect how Windows and applications behave.

2. IMPORTANCE / USAGE IN FORENSIC ANALYSIS

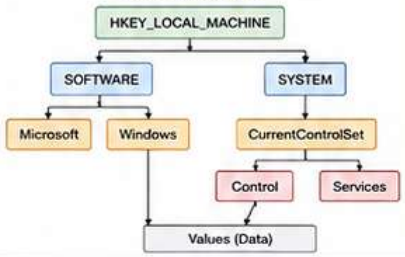
- ✓ **Stores user and system activity:** Contains traces of user actions, application usage, recent files, devices, and network connections.
- ✓ **Reveals system configuration:** OS settings, installed programs, services, and hardware information.
- ✓ **Tracks timeline events:** Timestamps (LastWrite, LastAccess) help build a timeline of activities.
- ✓ **Persists even after file deletion:** Registry can retain evidence even when files are deleted.
- ✓ **Useful in malware forensics:** Malware often adds keys/values for persistence, autorun, or configuration.
- ✓ **Helps in user profiling:** Stores user accounts, preferences, installed apps, history, and USB usage.
- ✓ **Critical for incident response:** Helps identify the cause, impact, and extent of an incident.

3. REGISTRY ORGANIZATION

The Windows Registry is a hierarchical tree structure consisting of:

- **HKEY (Root Keys)**
- **Sub Keys**
- **Values**

Hierarchy Example



4. FUNCTIONS OF REGISTRY HKEYs (ROOT KEYS)

HKEY (Root Key)	Full Form	Location / Scope	Function / Contains
HKEY_CURRENT_USER (HKCU)	Current User	Loaded from NTUSER.DAT of the currently logged-in user.	Stores user-specific settings such as desktop, Control Panel, environment variables, user preferences, and recent activities.
HKEY_LOCAL_MACHINE (HKLM)	Local Machine	Stored in system hives (SYSTEM, SOFTWARE, SAM, SECURITY).	Stores system-wide hardware, software, services, drivers, and system configuration.
HKEY_USERS (HKU)	Users	Contains all user profiles loaded on the system.	Includes all HKCU hives for every user. Useful to analyze other user accounts.
HKEY_CLASSES_ROOT (HKCR)	Classes Root	Merged view of HKLM\Software\Classes and HKCU\Software\Classes.	Stores file associations, COM objects, OLE, and shell information.
HKEY_CURRENT_CONFIG (HKCC)	Current Config	Represents the current hardware profile.	Stores current hardware profile information (similar to HKLM\System\CurrentControlSet\Hardware Profiles).

Note: HKCU, HKLM, HKCR, and HKCC are loaded when the system starts. HKU contains subkeys for each loaded user hive.

5. ANY SIX WINDOWS REGISTRY TERMINOLOGY

Term	Description
Hive	A top-level file that contains a portion of the Registry. Examples: SYSTEM, SOFTWARE, SAM, SECURITY, NTUSER.DAT.
Key	Similar to a folder in Windows Explorer. Keys can have sub keys and values.
Subkey	A key within another key. Forms the hierarchical structure.
Value	The actual data stored in a key. Each value has a name, type, and data.
Value Name	The name of a value under a key.
Value Type	Defines the type of data stored (e.g., REG_SZ, REG_DWORD, REG_BINARY, REG_QWORD, REG_MULTI_SZ, REG_EXPAND_SZ).
Data	The actual content or setting associated with a value.
Registry Hive File	Physical file on disk where registry data is stored (e.g., NTUSER.DAT, SYSTEM, SOFTWARE).
LastWrite Time	Timestamp of the last modification of a key or value.
Volatile Key	Dota that exists in memory only and not written to disk (e.g., certain system keys).

6. COMMON REGISTRY HIVE FILES AND THEIR LOCATION

Hive File	Description	Default Location
NTUSER.DAT	Stores HKCU (user-specific settings)	C:\Users\<User>\NTUSER.DAT
SYSTEM	Stores system configuration and hardware info	C:\Windows\System32\config\SYSTEM
SOFTWARE	Stores installed software and system settings	C:\Windows\System32\config\SOFTWARE
SAM	Stores user accounts and password hashes	C:\Windows\System32\config\SAM
SECURITY	Stores security policy and local security data	C:\Windows\System32\config\SECURITY
DEFAULT	Default user profile for new accounts	C:\Windows\System32\config\DEFAULT

These files can be extracted from disk images or live systems for forensic analysis using tools like Registry Editor (regedit), FTK Imager, RegRipper, Redline, etc.

7. WHAT CAN BE FOUND IN THE REGISTRY (FORENSIC ARTIFACTS)

- User account information and last logon details
- MRU (Most Recently Used) lists
- Recent documents and file history
- USB device history
- Installed programs and uninstall information
- Network connections and mapped drives
- Run keys (startup programs)
- System and application configuration
- Time zone, system time, and locale settings
- AutoRun and persistence mechanisms (malware)
- Browser and application traces
- Event log settings and services

Common Run Keys (Autorun Locations)
HKCU\Software\Microsoft\Windows\CurrentVersion\Run
HKLM\Software\Microsoft\Windows\CurrentVersion\Run
HKLM\Software\Wow6432Node\Microsoft\Windows\CurrentVersion\Run
HKCU\Software\Microsoft\Windows\CurrentVersion\RunOnce
HKLM\Software\Microsoft\Windows\CurrentVersion\RunOnce

Malware often uses Run/RunOnce keys for persistence.

8. COMMON REGISTRY VALUE TYPES

Value Type	Description	Example
REG_SZ	String (text) value	"C:\Program Files\App\app.exe"
REG_EXPAND_SZ	Expandable string	"%SystemRoot%\system32"
REG_DWORD	32-bit number	0x00000001 (1)
REG_QWORD	64-bit number	0x0000000000000001
REG_BINARY	Binary data	ff d8 ff e0 00 10 4a 46 49 46
REG_MULTI_SZ	Multiple strings	"txt\0docx\0pdf\0"

9. TOOLS AND COMMANDS

Tools

- Regedit (Built-in Registry Editor)
- RegRipper
- Redline
- Autopsy
- FTK Imager
- X-Ways Forensics
- Volatility (for memory registry analysis)

Common Commands

- **regedit** – Open Registry Editor
- **reg query** – Query registry keys/values
- **reg export** – Export registry key to .reg file
- **reg save** – Save hive to file
- **reg load** – Load hive from file

KEY TAKEAWAYS

- ✓ The Registry is a vital source of forensic evidence.
- ✓ It stores configuration, user activity, and system events.
- ✓ Artifacts in the Registry help build timelines and understand user behavior.
- ✓ Knowledge of hives, keys, values, and data types is essential.
- ✓ Always acquire and analyze Registry carefully and legally.

EXAM TIP

For 6 Marks:

1. Define Windows Registry.
2. Explain its importance in forensics.
3. Draw Registry organization (tree).
4. Describe functions of HKEYs.
5. List six Registry terminology with meaning.

IMPORTANT NOTE

Always maintain chain of custody.
Backup hives before editing.
Registry changes can affect the system.

14. SSD vs HDD / SSD FORENSIC CHALLENGES

HDD

SSD

Storage devices store digital data. Traditional **Hard Disk Drives (HDDs)** use magnetic spinning disks, while **Solid State Drives (SSDs)** use flash memory. Both behave very differently in forensic investigations.

1. ANY THREE BENEFITS OF SSD OVER HDD

1 Faster Performance
SSDs have much faster read/write speeds and lower access time, improving system boot, application loading and file transfers.

2 No Moving Parts (More Reliable)
SSDs have no mechanical components, so they are more durable, shock-resistant and consume less power.

3 Lower Power & Heat, Silent Operation
SSDs use less power, generate less heat and operate silently compared to HDDs.

2. WHY ARE SSDS A CHALLENGE FOR FORENSIC INVESTIGATORS?

- **TRIM Command**
  Deletes are not marked; data blocks are erased (or invalidated) immediately.
- **Garbage Collection**
  Unused blocks are cleaned in the background; deleted data is removed permanently.
- **Wear Leveling**
  Data is written to different physical locations to extend life; original locations are lost.
- **Over-Provisioning**
  Extra hidden space is used by SSD controller, not visible to OS or forensic tools.
- **Encryption & Secure Erase**
  Many SSDs use hardware encryption; Secure Erase can wipe all data instantly.
- **No Magnetic Residuals**
  HDDs may leave magnetic traces; SSDs do not, so advanced physical recovery is not possible.

3. HOW DOES SSD DIFFER FROM HDD? – OVERVIEW

	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Technology	Magnetic storage on spinning platters	Flash memory (NAND) chips
Moving Parts	Yes (platters, head, spindle motor)	No moving parts
Speed	Slower	Much faster
Durability	Less durable, prone to physical shock	More durable, shock-resistant
Power Consumption	Higher	Lower
Noise & Heat	Noisy, generates more heat	Silent, generates less heat
Access Time	Higher (ms range)	Very low (µs range)
Lifespan	Long (mechanically)	Limited write cycles (P/E cycles)
Typical Use	Bulk storage, archives	OS, applications, high performance

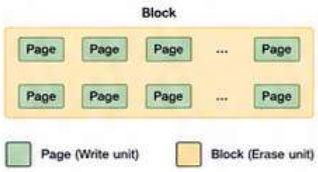
4. DETAILED COMPARISON: SSD vs HDD

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)	Explanation
1. Storage Technology	Magnetic recording on spinning platters	Electrical charge stored in NAND flash memory	HDD uses magnetic fields; SSD stores data in flash cells.
2. Components	Platters, spindle motor, read/write head, actuator arm, controller	NAND flash memory chips, controller, DRAM cache (optional)	HDD has mechanical parts; SSD is electronic.
3. Data Access	Head moves to correct track and sector	Direct electronic access to memory cells	No seek time in SSD; much faster random access.
4. Speed (Read/Write)	~100–200 MB/s (SATA)	~500 MB/s to 7000+ MB/s (SATA/NVMe)	SSDs are several times faster.
5. Data Transfer Rate	Limited by rotational speed (5400/7200 RPM)	Limited by interface (SATA, NVMe PCIe)	SSDs not limited by mechanical rotation.
6. Durability	Sensitive to shock and vibration	Shock-resistant, drop-resistant	No moving parts in SSD.
7. Noise	Produces noise (motor + seek)	Silent operation	SSDs are completely silent.
8. Power Consumption	Higher (5–10 W)	Lower (1–3 W)	Important for laptops and mobiles.
9. Heat Generation	Higher	Lower	SSDs run cooler.
10. Lifespan	Long mechanical life	Limited by write cycles (P/E cycles)	SSD cells wear out after many writes.
11. Wear Leveling	Not applicable	Used to distribute writes evenly across cells	Extends SSD life but changes data locations.
12. TRIM Support	Not applicable	Yes (enables OS to tell SSD what is deleted)	TRIM causes deleted data to be removed.
13. Data Recovery	Deleted data often recoverable	Deleted data usually unrecoverable	Due to TRIM, GC, wear leveling, encryption.
14. Capacity	Higher (up to 20TB+)	Lower to moderate (up to 8TB consumer)	HDDs still cheaper for very large storage.
15. Cost per GB	Lower	Higher	SSDs are costlier per GB.
16. Interface	SATA, SAS	SATA, NVMe (PCIe), SAS	NVMe SSDs are much faster.
17. Typical Use	Mass storage, backups, archives	OS, applications, database, high performance	SSDs preferred for speed & reliability.

5. HOW SSD WORKS (Key Concepts Affecting Forensics)

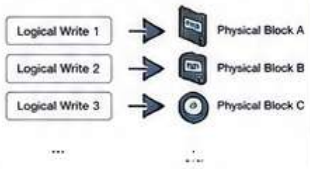
NAND Flash Structure

Data is stored in pages (e.g., 4KB, 8KB)
Pages are grouped into blocks (e.g., 128–512 pages)
Blocks are the smallest unit that can be erased



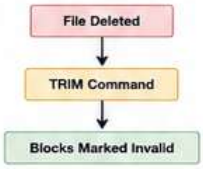
Wear Leveling

To extend life, controller writes data to different physical blocks so that wear is evenly distributed. This changes the physical location over time.



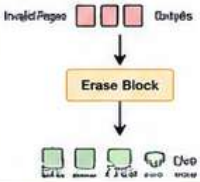
TRIM Command

When a file is deleted, OS sends TRIM to SSD telling which blocks are no longer in use. SSD marks them as invalid.



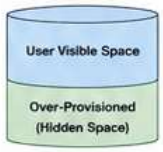
Garbage Collection (GC)

Background process that removes invalid pages/blocks and erases entire blocks so they can be reused.



Over-Provisioning

Extra hidden space is reserved by SSD controller to improve performance and manage GC.



6. ANY SIX WINDOWS REGISTRY TERMINOLOGY

Term	Meaning
1. **Hive**	A top-level file that stores the Registry data (e.g., SYSTEM, SOFTWARE, SAM).
2. **Key**	A container in the Registry that can have subkeys and values (like a folder).
3. **Subkey**	A key inside another key (forms a hierarchical structure).
4. **Value**	The actual data stored in a key (e.g., settings, paths, numbers).
5. **Value Name**	The name of a value (e.g., Start, Path, Type).
6. **Data Type**	Type of data stored (e.g., REG_SZ, REG_DWORD, REG_BINARY, REG_MULTI_SZ).
7. **Data**	The content of the value according to its data type.
8. **LastWrite Time**	Timestamp of the last change to a key or value.
9. **Default Value**	A value with no name (shown as "(Default)").
10. **Volatile Key**	Loaded in memory only; not written to disk (e.g., certain system keys).

7. MAJOR WINDOWS REGISTRY HKEYs (ROOT KEYS) AND THEIR FUNCTIONS

Hive (HKEY)	Full Name	Main Function / Contains
HKEY_CURRENT_USER	HKCU	Settings for the currently logged-in user (preferences, desktop, environment, recent items, software settings).
HKEY_LOCAL_MACHINE	HKLM	System-wide configuration (hardware, software, services, SAM, SECURITY, SYSTEM hives).
HKEY_USERS	HKU	All user profiles loaded on the system.
HKEY_CLASSES_ROOT	HKCR	File associations, COM objects, shell extensions, and system classes.
HKEY_CURRENT_CONFIG	HKCC	Current hardware profile configuration.

ℹ HKLM contains subkeys: SYSTEM, SOFTWARE, SAM, SECURITY, DEFAULT.

8. FORENSIC ARTIFACTS FOUND IN REGISTRY

- User activity (RunMRU, RecentDocs, TypedURLs)
- Installed programs and uninstall information
- USB device history
- Network connections and mapped drives
- System and application configuration
- Prefetch, services, startup programs
- Time zone, clock settings
- User accounts and logon information
- Event log settings
- MRU lists (Open/Save)



9. KEY TAKEAWAYS

- SSDs are faster, more reliable and power efficient than HDDs.
- SSDs use wear leveling, TRIM and garbage collection which make deleted-data recovery extremely difficult.
- HDDs are more forensic-friendly as deleted data can often be recovered.
- Investigators must use specialized tools and imaging techniques for SSDs.

10. EXAM TIP

For 6 Marks:

1. List 3 benefits of SSD over HDD.
2. Explain why SSDs are challenging.
3. Give differences between SSD and HDD.
4. Add key concepts: Wear leveling, TRIM, GC and deleted-data challenges.

15. BITLOCKER

BitLocker is a full-volume encryption feature by Microsoft that helps protect data by encrypting entire drives. It prevents unauthorized access if a drive is lost, stolen, or the system is accessed offline.

1. WINDOWS VERSIONS THAT SUPPORT BITLOCKER

Windows Edition / Version	BitLocker Support
Windows 10 / 11 Pro	✅ Yes
Windows 10 / 11 Enterprise	✅ Yes
Windows 10 / 11 Education	✅ Yes
Windows 10 / 11 Home	❌ No
Windows 7 Ultimate / Enterprise	✅ Yes
Windows 8 / 8.1 Pro / Enterprise	✅ Yes
Windows 7 Professional / Home Premium	❌ No

ℹ Note: BitLocker is not available in Windows Home editions.

2. HARDWARE REQUIREMENTS

- Trusted Platform Module (TPM) 1.2 or 2.0 (TPM 2.0 is recommended)
- UEFI firmware (with Secure Boot capability recommended)
- A compatible system BIOS
- Sufficient free disk space for encryption (at least 1.5 times the size of the largest file on the drive)
- Removable USB drive for storing recovery key (recommended)
- A password or PIN for user authentication

TPM 2.0

3. SOFTWARE REQUIREMENTS

- Supported Windows edition (as listed in the left table)
- NTFS file system
- Administrative privileges
- TPM driver enabled in BIOS/UEFI
- Windows Management Instrumentation (WMI) service must be running
- For system drive encryption, Secure Boot is highly recommended
- For extra security: Windows updates enabled

4. HOW BITLOCKER WORKS (OVERVIEW)

Authentication (Password / PIN / TPM) → Encrypted Drive (Full Volume)

- BitLocker encrypts the entire volume (OS, files, free space).
- Uses AES encryption (128-bit or 256-bit).
- The encryption key is sealed (protected) using TPM and stored in system.
- At startup, BitLocker verifies system integrity and user authentication before unlocking the drive.
- Recovery Key is generated and can be saved to Microsoft account, USB, file, or printed.

Types of BitLocker Protection

TPM Only
Uses TPM to protect the key.

TPM + PIN
Requires TPM and a PIN at startup.

TPM + Startup Key
Requires a USB key at startup.

Password Only
Uses a password (no TPM).

5. ONE COMPUTER HARDWARE ENCRYPTION PROGRAM

Example: BitLocker Drive Encryption (Microsoft)

BitLocker is a hardware-based full disk encryption solution built into Windows.

Explanation:

- Works closely with TPM (hardware security chip) to store encryption keys securely.
- Encrypts entire drives (operating system drive, fixed data drives, removable drives).
- Provides protection against offline attacks and data theft.
- Uses strong AES-128/256 encryption.
- Supports pre-boot authentication to prevent unauthorized access.
- Integrates with Active Directory for recovery key backup in enterprise environments.

ℹ **Other hardware encryption programs:**
PGP Whole Disk Encryption, VeraCrypt (with hardware acceleration), McAfee Drive Encryption.

6. USAGE OF WINDOWS REGISTRY AND BITLOCKER IN FORENSIC ANALYSIS

A. WINDOWS REGISTRY IN FORENSIC ANALYSIS

The Windows Registry stores configuration, user activity, system settings, and application data. It is a crucial source of digital evidence.

Registry Key (Hive)	Forensic Significance
HKCU (HKEY_CURRENT_USER)	User activities, recent documents, programs run, USB usage, user preferences, MRU lists.
HKLM (HKEY_LOCAL_MACHINE)	System configuration, installed programs, hardware info, services, network settings.
SYSTEM	Boot and shutdown times, driver details, control sets.
SOFTWARE	Installed applications, software settings, security applications, policies.
SAM	User accounts, password hashes (protected).
SECURITY	Security policies, audit logs, LSA secrets.

Forensics Use:

- Timeline creation, user behavior analysis, program execution tracking, USB and device history, system events, security logs, etc.

B. BITLOCKER IN FORENSIC ANALYSIS

BitLocker protects data by encrypting entire drives, making forensic analysis challenging without proper keys.

Forensic Considerations:

- If drive is locked and keys are unavailable, data cannot be accessed.
- Investigators must acquire the recovery key, password, PIN, or TPM (in some cases).
- BitLocker metadata and configuration details are stored in Windows Registry.
- Evidence of BitLocker usage itself can be valuable (e.g., whether encryption was enabled, when it was disabled).
- Live system acquisition may be required if the drive is unlocked.

Important Registry locations related to BitLocker:

- HKLM\SOFTWARE\Policies\Microsoft\FVE – BitLocker policy settings
- HKLM\SYSTEM\CurrentControlSet\Control\BitLocker – Status and settings
- HKLM\SOFTWARE\Microsoft\BitLocker – Drive encryption info
- HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2 – Encrypted drive history

7. BITLOCKER REGISTRY ARTIFACTS (EXAMPLES)

Registry Path	Description / Forensic Value
HKLM\SOFTWARE\Microsoft\FVE\Volatile	BitLocker status, encryption method, protection enabled, auto-unlock info.
HKLM\SOFTWARE\Policies\Microsoft\FVE	Group policy settings for BitLocker.
HKLM\SYSTEM\CurrentControlSet\Control\BitLocker	BitLocker driver and configuration.
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon	Auto-unlock related information.
HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Setup\RecoveryConsole	Recovery environment settings.

ℹ These artifacts help investigators determine if BitLocker was used, when it was enabled/disabled, and what recovery options exist.

8. SUMMARY – BITLOCKER IN FORENSICS

- BitLocker is a full-disk encryption solution in Windows.
- Supported in Windows Pro, Enterprise, Education editions.
- Requires TPM, UEFI, and NTFS file system.
- Protects data using strong AES encryption.
- Registry stores BitLocker configuration and status information.
- Without keys, encrypted drives are extremely difficult to analyse.
- Acquiring keys or performing live acquisition is critical.
- Knowledge of Registry + BitLocker artifacts improves forensic outcomes.

BitLocker

9. KEY TAKEAWAYS

- ✓ BitLocker protects entire drives from offline attacks.
- ✓ Hardware (TPM) + software (Windows) work together.
- ✓ Registry is essential for finding evidence of encryption and user activity.
- ✓ Always try to obtain recovery keys and perform live acquisition if possible.

10. EXAM TIP

For 6 Marks:

1. List supported Windows versions.
2. Write hardware and software requirements.
3. Explain one hardware encryption program (BitLocker).
4. Explain Registry usage in forensics.
5. Explain BitLocker usage and forensic challenges.
6. Use diagrams/table for better presentation.

11. IMPORTANT NOTE

- Always maintain chain of custody.
- Do not modify registry or encrypted drives.
- Use forensic tools (FTK, EnCase, RegRipper, Magnet AXIOM) to extract artifacts.
- Legal authorization may be required to access encrypted data.

16. BOOT SEQUENCE ★★★★

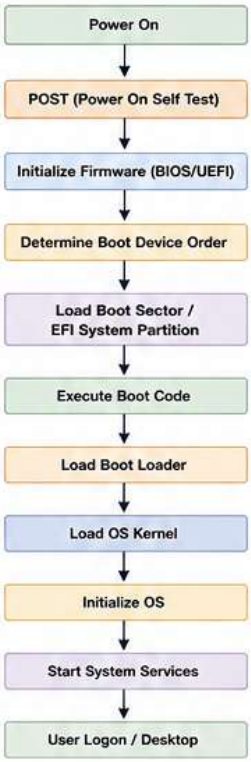
Explain the operations/steps involved in the boot sequence.

Boot sequence is the process by which a computer is started and the operating system is loaded into memory after the power is turned on or the system is restarted.

BOOT SEQUENCE – OPERATIONS / STEPS (EXACT SEQUENCE)

1	**Power On**	The user presses the power button. The system receives power and the CPU starts execution from a fixed location.
2	**POST (Power On Self Test)**	BIOS/UEFI performs POST to check hardware components (CPU, RAM, keyboard, storage, etc.). If any critical error is found, the system halts.
3	**Initialize Firmware**	BIOS/UEFI initializes and configures system hardware and devices. It sets up chipset, memory controller, PCI devices, etc.
4	**Determine Boot Device Order**	Firmware reads the boot order from BIOS/UEFI settings. Devices are checked in the configured order (e.g., HDD, SSD, USB, CD/DVD, Network).
5	**Load Boot Sector / EFI System Partition**	The firmware loads the boot sector (MBR/VBR) from the selected device (Legacy BIOS) or the EFI System Partition (UEFI).
6	**Execute Boot Code**	Boot code in the boot sector is executed. It locates and loads the next stage boot loader.
7	**Load Boot Loader**	Boot loader (e.g., Windows Boot Manager / bootmgr / GRUB) is loaded into memory. It provides boot options (if any) and loads the OS kernel.
8	**Load OS Kernel**	Boot loader loads the operating system kernel (e.g., ntoskrnl.exe in Windows) into memory.
9	**Initialize OS**	OS kernel initializes system drivers, services, and hardware abstractions.
10	**Start System Services**	OS starts essential services and system processes.
11	**User Logon / Desktop**	Finally, the system is ready and the user logon screen or desktop is displayed.

BOOT FLOW DIAGRAM



BIOS vs UEFI (Brief)

BIOS (Legacy)	UEFI
Runs in real mode	Runs in protected mode
Uses MBR partition	Uses GPT partition
Boot code from MBR	Boot from EFI System Partition (ESP)
Limited features	More secure, faster, supports large disks
No Secure Boot	Supports Secure Boot

IMPORTANT POINTS

- ✓ Boot sequence starts from firmware and ends with the OS ready for user.
- ✓ The boot order decides which device will be attempted first.
- ✓ If the OS is not found, the system may show an error like "Operating System not found".
- ✓ Both BIOS and UEFI perform the same purpose but in different ways.

COMMON FORENSIC USE

Boot artifacts help in analyzing:

- Last boot time
- Boot device used
- Failed boot attempts
- Boot configuration changes
- User activity timeline

KEY TAKEAWAY:
Boot sequence is a chain of operations that starts from power on, hardware check, boot device selection, OS loading and finally the system ready for use.

EXAM TIP
For 4 Marks: Write the 11 steps in order with brief explanation or draw the flow diagram.

17. DELETED NTFS / RECYCLE BIN ★★★★

What happens when someone empties the Recycle Bin in NTFS?

In NTFS, deleting a file (or emptying Recycle Bin) does not immediately erase the actual data.
It only removes references to the file from the file system.

WHAT HAPPENS WHEN RECYCLE BIN IS EMPTIED IN NTFS?

1 File is Permanently Deleted (Logically)
When a file in Recycle Bin is emptied, NTFS considers the file permanently deleted. The file is no longer visible in the file system.

2 MFT Entry is Updated
The file's MFT (Master File Table) entry is changed.
- The file name and path are removed.
- The $STANDARD_INFORMATION attribute is updated.
- The IN_USE flag in the MFT entry is cleared (marked as free).

3 Space Marked as Available
The clusters (disk space) that were occupied by the file are marked as available in the file system.
- In the $Bitmap, those clusters are cleared (0) to indicate free space.
- The space can now be used for new data.

4 File Data May Still Remain
The actual file data (contents) remains on the disk until it is overwritten by new data.
- NTFS does not automatically erase or wipe the data.
- Only the references (MFT entry, directory entry) are removed.

5 Recovery Possibilities
As long as the file data is not overwritten:
- The file can be recovered using forensic tools.
- Tools analyze MFT, $Bitmap, and file data signatures.
- Success depends on how much new data has been written.

NTFS DELETION PROCESS (WHEN RECYCLE BIN IS EMPTIED)



WHAT CHANGES IN NTFS?

Component	What Happens
MFT Entry	Entry is not deleted, but marked as free (IN_USE flag = 0).
File Name	Removed from directory index (no longer visible).
Allocation (Clusters)	Clusters are marked free in $Bitmap.
File Data	Actual data remains on disk until overwritten.
Indexes ($I30)	Directory index entry for the file is removed.

FORENSIC RECOVERY POSSIBILITIES

- Data can be recovered using tools like: Recuva, TestDisk, Photorec, FTK Imager, EnCase, Autopsy, etc.
- Recovery success depends on:
  - How much new data has been written.
  - Type of file, fragmentation, cluster size.
  - Whether TRIM (in SSD) or overwriting has occurred.



KEY POINTS TO REMEMBER

- ✓ Emptying Recycle Bin = removing references, not data.
- ✓ MFT entry is updated and marked free.
- ✓ Disk space is marked available in $Bitmap.
- ✓ Data may remain until overwritten.
- ✓ Files can be recovered using forensic techniques.

EXAM TIP
For 4 Marks:
Write the 5 points (MFT update, space available, data remains, recovery possible) with a neat diagram.

IMPORTANT NOTE

- On SSDs with TRIM enabled, deleted data may be erased by the controller (garbage collection). So recovery chances are lower compared to HDDs.

On HDDs, chances of recovery are higher.

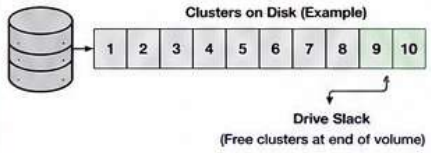
18. SLACK SPACE ★★★

Slack space is unused space that exists in storage areas (disk or memory) but may still contain residual data. It can hold remnants of previously stored information and is useful in forensic analysis.

1. DRIVE SLACK (VOLUME SLACK)

- Occurs in the last cluster(s) or sector(s) of a partition/volume that are not used for any file.
- The file system marks this space as free, but old data may still remain.
- Also called Volume Slack or Unallocated Slack.

Diagram:



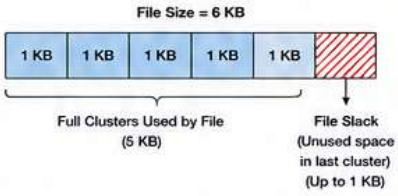
Key Points

- Present at the end of the volume.
- Not allocated to any file.
- Data may remain until overwritten.
- Useful for recovering deleted data fragments.

2. FILE SLACK (INTERNAL SLACK)

- Occurs in the last cluster allocated to a file when the file size does not completely fill that cluster.
- The unused portion of that cluster is slack space.
- Also called Internal Slack.

Diagram:



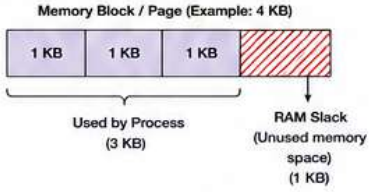
Key Points

- Exists within the last allocated cluster of a file.
- Not visible in file size.
- May contain remnants of old data.
- Recoverable using forensic tools.

3. RAM SLACK (MEMORY SLACK)

- Occurs in **RAM pages** or memory blocks that are allocated to processes but not completely used.
- The unused portion of the memory block is slack.
- **Data may persist** until the memory is reused.
- Important in memory forensics.

Diagram:



Key Points

- Unused portion of allocated memory.
- May contain sensitive remnants (passwords, keys, data fragments).
- Captured in memory dumps.

COMPARISON SUMMARY

Type	Where It Occurs	Related To	Visibility	Forensic Value
Drive Slack	End of volume/partition	File System (Disk)	Hidden (Unallocated)	Moderate – May contain deleted data
File Slack	Last cluster of a file	Individual File	Hidden (Within file)	High – Often contains data remnants
RAM Slack	Unused memory in allocated block/page	Process / Memory	Hidden (In Memory)	High – Contains sensitive artifacts

IMPORTANCE IN FORENSICS

- Slack space can hold remnants of deleted files, fragments of documents, passwords, encryption keys, images, etc.
- Not referenced by file system, thus remains hidden from normal users.
- Forensic tools (e.g., FTK Imager, EnCase, Autopsy, Sleuth Kit) can extract and analyze slack space.

EXAM TIP

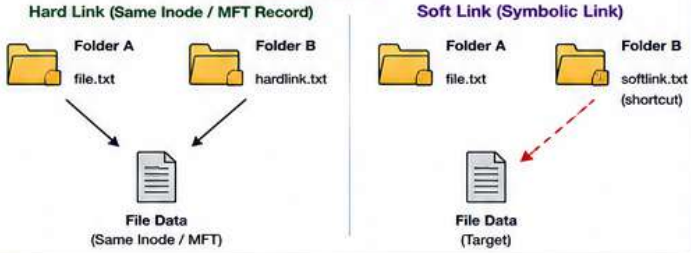
Remember the Location: Drive Slack → End of Volume File Slack → End of file's Last Cluster RAM Slack → End of Memory Block/Page

19. HARD LINK vs SOFT LINK ★★★★

Links allow a file to have more than one name or reference.
Hard Link points directly to the file data (same inode/record). Soft Link (Symbolic Link) points to the path of the target file.

Feature	HARD LINK	SOFT LINK (SYMBOLIC LINK)
Definition	A reference to the actual file data by sharing the same inode (UNIX) or MFT record (NTFS).	A special file that contains the path to the target file or directory.
File System Support	NTFS, EXT2/3/4, HFS+	NTFS, FAT32, exFAT, EXT2/3/4, HFS+, etc.
Link Creation	Not a separate file; additional directory entry for the same file.	A separate file that points to the target path.
Inode / MFT Record	Same inode / MFT record as the original file.	Different inode / MFT record.
Data Storage	No additional space used; only another directory entry.	Uses space to store the path of the target.
Cross File System	Not allowed.	Allowed (can point to another file system).
Supports Directories	Generally not allowed (except in some UNIX with restrictions).	Yes, can link to files and directories.
If Original File is Deleted	Data remains; accessible via other hard links.	Link is broken (dangling link); cannot access data.
If Link is Deleted	Other links remain; file still accessible.	Link is removed; original file not affected.
Identification	Both links show same inode number (UNIX) or MFT reference (NTFS).	Different inode number; file type usually 'L' (ln -l in UNIX).
Example (UNIX)	`ln file.txt hardlink.txt`	`ln -s file.txt softlink.txt`
Example (Windows NTFS)	`mklink /H hardlink.txt file.txt`	`mklink /D dirlink folder` (for directory) `mklink softlink.txt file.txt` (for file)
Use Cases	Backup, system files, save space while keeping multiple names.	Shortcuts, references to files/directories, cross-volume links.

VISUAL DIAGRAM



KEY DIFFERENCES AT A GLANCE

- Hard link = another name for the same file.
  Soft link = shortcut to the file path.
- Hard link cannot cross file systems;
  Soft link can cross file systems.
- Hard link survives deletion of original (until last link removed);
  Soft link breaks if original is deleted or moved.
- Hard link does not use extra space;
  Soft link uses space to store path.

EXAM TIP

Remember:

1. Hard Link → Same data (same inode), not allowed across file systems.
2. Soft Link → Path reference, works across file systems, can break.

IMPORTANT NOTE

On Windows, run Command Prompt as Administrator to create links.
On UNIX/Linux, use: ln (hard link) and ln -s (symbolic link).

20. DELETED-DATA RECOVERY CHALLENGES ★★★

When a file is deleted, it is not immediately removed from the storage device.
However, several factors make recovery difficult or sometimes impossible.

1. WHAT HAPPENS WHEN A FILE IS DELETED?

File Exists → File Deleted → Space Marked as Available → Data May Remain Until Overwritten

- The file's directory entry (e.g., MFT entry in NTFS) is marked as deleted.
- The space previously used by the file is marked as free/available.
- The actual data usually remains on the disk until it is overwritten.
- Recovery is possible only if the data has not been overwritten.

2. MAJOR CHALLENGES IN DELETED-DATA RECOVERY

Data Overwriting	New data written to the same location overwrites old data, making recovery impossible.
File System Activity	File system metadata (e.g., MFT, FAT, allocation tables) may be updated, reusing clusters and altering evidence.
TRIM / Garbage Collection (SSDs)	SSDs use TRIM and Garbage Collection which erase deleted data blocks proactively, making recovery very difficult.
Wear Leveling (SSDs)	Data is distributed across the flash memory to balance wear, so original location of data is unknown.
Encryption	Full-disk or file encryption makes raw data unreadable without decryption keys.
Fragmentation	Files are stored in fragments; some parts may be overwritten while other parts remain, leading to partial recovery.
Time Factor	More time = more system activity = higher chance of overwrite and lower chance of recovery.
Limited Metadata	If file system metadata is damaged or overwritten, file name, size, timestamps, and structure may be lost.
User Behavior	Actions like formatting, defragmentation, OS updates, cleaning tools, or secure erase make recovery hard.

3. FACTORS THAT REDUCE RECOVERY CHANCES

- Saving new files to the same drive
- Formatting (Quick or Full)
- Running Disk Cleanup
- Defragmentation (HDD)
- TRIM (SSD)
- Garbage Collection (SSD)
- Reinstalling Operating System
- Increasing system usage
- Secure wipe / Erase utilities
- Physical damage to storage
- Encryption without keys
- RAID rebuild or synchronization

Condition	HDD (Traditional)	SSD (Solid State Drive)	Recovery Possibility
File deleted, not overwritten	High	Low to Moderate	High
Partially overwritten	Moderate (partial recovery)	Low	Moderate to Low
After formatting (quick)	Moderate	Very Low	Low
After formatting (full)	Very Low	Very Low	Very Low
After TRIM / GC (SSD)	Not Applicable	Very Low to None	Very Low / None
After secure wipe / encryption	None	None	None

4. FORENSIC BEST PRACTICES TO IMPROVE RECOVERY

- ✓ Immediately stop using the storage device.
- ✓ Acquire a forensic image (bit-stream copy) before analysis.
- ✓ Work on the image, never on the original evidence.
- ✓ Use write blockers to prevent changes.
- ✓ Use suitable tools (e.g., EnCase, FTK, R-Studio).
- ✓ Recover to a different drive.
- ✓ Document every step and maintain chain of custody.

Golden Rule: The sooner the data is acquired, the higher the chances of recovery.

6. KEY TAKEAWAYS

- Deleted data is not guarnateed to be recoverable.
- The biggest enemy of recovery is OVERWRITE.
- SSDs present more challenges than HDDs.
- Time, system activity, and user actions reduce recovery chances.
- Proper handling and imaging increase the probability of successful recovery.

21. MICROSOFT WINDOWS STARTUP TASKS ★★★★

When a Windows system is turned on, a series of startup tasks are performed before the user logon screen appears.
These tasks are critical for initializing hardware, loading the OS, and starting essential services.

1. WINDOWS STARTUP PROCESS – OVERVIEW

1	**Power On**	System power is supplied and hardware initialization begins (BIOS/UEFI).
2	**POST (Power On Self Test)**	BIOS/UEFI performs hardware tests (CPU, RAM, storage, keyboard, etc.). If any critical error is found, boot stops.
3	**BIOS / UEFI Initialization**	BIOS/UEFI initializes devices, configures chipset, memory controller, and PCI devices.
4	**Boot Device Selection**	BIOS/UEFI reads the boot order and selects the boot device (e.g., HDD, SSD, USB, CD/DVD, Network).
5	**Load Boot Sector / EFI System Partition**	MBR (Legacy BIOS) or EFI System Partition (UEFI) is loaded into memory.
6	**Execute Boot Code**	Boot code from boot sector / EFI partition is executed. It loads the boot loader.
7	**Load Boot Loader**	Boot loader (e.g., Windows Boot Manager) loads the Windows kernel (ntoskrnl.exe) into memory.
8	**Load Windows Kernel**	Kernel initializes core components and starts hardware abstraction layer.
9	**Initialize System Services & Drivers**	Windows initializes system drivers, services, and hardware devices (Plug and Play).
10	**User Logon / Desktop**	Logon screen appears. After successful authentication, the user profile loads and the desktop is displayed.

2. DETAILED STARTUP TASKS AFTER KERNEL LOAD

Order	Startup Task	Description
1	**Hardware Detection**	Detects and configures Plug and Play (PnP) devices.
2	**Load Device Drivers**	Loads necessary drivers for system and hardware devices.
3	**Start System Services**	Starts critical Windows services (e.g., RPC, Event Log, Plug and Play).
4	**Mount File Systems**	Mounts system volumes and file systems (NTFS, FAT, etc.).
5	**Registry Initialization**	Loads system registry hives (SYSTEM, SOFTWARE, SAM, SECURITY, DEFAULT).
6	**Winlogon Start**	Winlogon.exe starts (handles secure logon process).
7	**Group Policy Processing**	Applies Local Group Policy settings.
8	**User Profile Service**	Loads user profile and environment settings.
9	**Startup Programs**	Runs startup programs (registry: Run keys, Startup folders).
10	**Network Initialization**	Initializes network connections and services.
11	**Windows Services (Auto)**	Starts remaining automatic services.
12.	**User Interface**	Explorer.exe starts and desktop is displayed.

3. IMPORTANT LOCATIONS (ARTIFACTS)

- ✓ **Registry Run Keys:** HKLM\...\Run, HKCU\...\Run, RunOnce
- ✓ **Startup Folder:** %ProgramData%\Microsoft\Windows\Start Menu\Programs\Startup
- ✓ **Services:** HKLM\SYSTEM\CurrentControlSet\Services
- ✓ **Event Logs:** System log (Event ID 6005 – Event log started)
- ✓ **Task Scheduler:** Startup-linked tasks

4. IMPORTANCE IN FORENSIC ANALYSIS

- Helps identify persistence mechanisms (malware, backdoors).
- Startup entries can indicate unauthorized or suspicious programs.
- Artifacts reveal user activity, time of execution, and system changes.
- Artifacts are found in Registry, Event Logs, Startup folders, Prefetch, Amcache.

EXAM TIP

Remember the flow: Power On → POST → BIOS/UEFI → Boot Device → Boot Sector/ESP → Boot Code → Boot Loader → Kernel → Drivers/Services → Logon.
Focus on artifacts: Registry Run Keys, Startup Folder, Services, Event Logs.

1. ACQUISITION TOOLS AND METHODS ★★★

Acquisition is the process of collecting digital evidence from a source device in a forensically sound manner without altering the original data.

1.1 COMMON ACQUISITION TOOLS

Tool	Description
FTK Imager	Creates bit-by-bit forensic images of drives and partitions. Supports hashing (MD5, SHA-1), compression and verification.
EnCase Forensic Imager	Forensic imaging and data capture tool with write-blocker support and hashing.
dd (Linux)	Command-line utility to create bit-stream images. Example: `dd if=/dev/sda of=image.dd bs=4M conv=noerror,sync`
Guymager	Linux based GUI tool for acquiring images from drives and partitions.
ddrescue (Linux)	Acquiree data from failing media by skipping bad sectors and retrying later.
KAPE (Kroll Artifact Parser & Extractor)	Forensic collection tool that targets artifacts from Windows systems (Registry, Event Logs, Browser data, etc.).
Live Response Tools (Helix, Velociraptor)	Collect data from live systems without full imaging (memory, running processes, logs).
Hardware Write Blockers (Tableau, WiebeTech)	Hardware devices placed between the source and destination to prevent any write operation to the original media.

1.2 ACQUISITION METHODS

1. **Physical Acquisition (Bit-by-Bit Imaging)**
   - Creates an exact copy of the entire storage device or partition.
   - Includes allocated, unallocated, slack space, and deleted data.
2. **Logical Acquisition**
   - Copies only accessible files and folders using OS commands.
   - Faster but may miss deleted/hidden data.
3. **Live Acquisition**
   - Collects data from a running system (memory, network connections, processes).
   - Useful for volatile evidence.
4. **Remote Acquisition**
   - Collects data from remote systems over the network.
   - Used in cloud or corporate environments.
5. **Targeted Acquisition**
   - Collect specific data sets (emails, logs, Registry hives, browser data).
   - Minimizes data volume and time.

1.3 BEST PRACTICES FOR ACQUISITION

- Follow proper legal authorization.
- Document every step of the acquisition process.
- Use write blockers to protect original data.
- Verify the integrity of the acquired image using hash values.
- Store the image in a secure, sterile medium.
- Label the evidence properly.
- Maintain chain of custody.
- Use trusted tools and methods.

TYPES OF ACQUIRED DATA

- Disk / Partition Images
- Memory Dumps (RAM)
- System / Application Logs
- Registry Hives
- Network Data
- Mobile Device Data
- Cloud Data

KEY TAKEAWAY

The goal of acquisition is to obtain a forensically sound copy of data while preserving its integrity, authenticity and admissibility in court.

2. DIGITAL EVIDENCE VALIDATION METHODS / TOOLS ★★★

Validation ensures that the acquired digital evidence is an exact and unaltered copy of the original source.

2.1 VALIDATION METHODS

1. **Hash Value Verification**
   Generate hash (MD5, SHA-1, SHA-256) of the source and the acquired image. If both match, the image is an exact copy.
2. **Bit-by-Bit Comparison**
   Compare the original media and the acquired image bit-by-bit to ensure identical data.
3. **Re-imaging & Re-hashing**
   Re-acquire the image and generate hash again. If the hash matches the previous image, it confirms consistency.
4. **Spot Checking**
   Randomly verify files and folders from the image with the source for accuracy.
5. **Write Blocker Check**
   Ensure write blocker is functioning properly before acquisition to prevent data modification.

2.2 VALIDATION TOOLS

Tool	Purpose / Features
FTK Imager	Generates and verifies hash values (MD5, SHA-1, SHA-256). Performs image verification.
HashCalc	Free tool to compute hash values of files (MD5, SHA-1, SHA-256, etc.).
FCIV (Microsoft)	File Checksum Integrity Verifier from Microsoft Sysinternals.
md5deep (Linux)	Creates and verifies hashes of files and directories.
cmp (Linux)	Compares two files byte by byte.
ddrescue (Map File)	Verifies and resumes acquisition using map file to ensure data integrity.

2.3 HASH ALGORITHMS COMMONLY USED

Algorithm	Hash Size	Use
MD5	128-bit	Legacy use (not collision resistant)
SHA-1	160-bit	Legacy use (not collision resistant)
SHA-256	256-bit	Recommended (secure)
SHA-512	512-bit	Highly secure

VALIDATION WORKFLOW

Source Device → Acquire (Image) → Generate Hash → Verify (Hash Match)

IMPORTANT POINTS

- Always validate immediately after acquisition.
- Document hash values in the case notes.
- Any change in hash value indicates alteration or corruption.
- Use strong hash algorithms (SHA-256 or above).

EXAMPLE

Source Drive SHA-256 : 3f1c2d4e5a6b7c8d9e0f112233445566778899aabbccddeeff0011223344556677
Image File SHA-256 : 3f1c2d4e5a6b7c8d9e0f112233445566778899aabbccddeeff0011223344556677
Result : **MATCH** → Image is an exact copy.

3. EVIDENCE RETENTION ★★★

Evidence retention is the process of securely storing digital evidence for the required period to ensure its availability, integrity and admissibility in legal proceedings.

3.1 PURPOSE OF EVIDENCE RETENTION

- Preserve the evidence in its original state.
- Prevent loss, damage or unauthorized access.
- Maintain integrity and authenticity.
- Ensure availability for future analysis or court.
- Comply with legal and organizational policies.

3.2 FACTORS AFFECTING RETENTION

- Type of case and legal requirements
- Organizational / company policies
- Court orders and jurisdiction laws
- Storage media life and environmental conditions
- Retention schedule / policy

3.3 BEST PRACTICES FOR EVIDENCE RETENTION

- Store in a secure, access-controlled evidence repository.
- Use original formats (bit-stream images) for retention.
- Maintain proper documentation and metadata.
- Use tamper-evident seals and secure containers.
- Regularly verify integrity (hash check) during retention.
- Maintain chain of custody logs.
- Follow retention schedule and legal hold instructions.
- Dispose of evidence securely after retention period (if permitted).

3.4 RETENTION STORAGE GUIDELINES

Guideline	Description
Storage Media	Use reliable media (Forensic HDD, WORM, Blu-ray, Secure NAS).
Environment	Cool, dry, dust-free, away from magnetic fields and direct sunlight.
Access Control	Limit access to authorized personnel only.
Backup	Maintain redundant copies in secure location.
Monitoring	Regular integrity checks and media health monitoring.
Documentation	Record storage location, media ID, hash values, date and responsible person.

KEY TAKEAWAY

Proper acquisition, validation and retention ensure that digital evidence remains reliable, untampered and admissible in court.

Remember: Acquire Carefully → Validate Always → Retain Securely